

OMG Business Architecture Core Metamodel

(Initial submission)

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Table of Contents

1	Introduction	4
1.1	Submitting Companies.....	4
1.2	Supporting organization	4
1.3	Status of the Document.....	4
1.4	Guide to the Submission.....	4
1.5	Proof of Concept.....	4
1.6	Submission Contacts.....	4
1.7	Change History.....	4
2	Problem to be solved	5
2.1	Building Block Pattern	6
2.2	Temporal aspects.....	7
2.3	Purposeful Systems and Capabilities	9
2.4	From system to the enterprise	13
2.5	Gap analysis and Transformation	14
2.6	BACM concept GRID	15
3	BACM & Existing Standards.....	16
3.1	BACM and UML 2.0 & MOF	16
3.2	BACM and BPMN	16
3.3	BACM and BMM	16
3.4	BACM and UAF	16
3.5	BACM and Business Model Canvas.....	16
3.6	BACM and Value Proposition Canvas	17
4	Specification	18
4.1	Overview.....	18
4.2	Notation convention.....	19
4.3	Upper Ontology & EA Patterns	19
4.4	Business Capabilities.....	31

4.5	Service & Product Model.....	32
4.6	Business Operating Model.....	32
4.7	Value Stream Model.....	35
4.8	Resource Operating Model.....	37
4.9	Resource Functionality Models	38
4.10	Organization Model.....	40
4.11	Information Model	41
4.12	Strategy Model	42
4.13	Transformation Model.....	47
5	Compliance Levels.....	49
5.1	PIM Compliance.....	49
5.2	Other?.....	49
Appendix A: Glossary.....		50
5.3	The BACM GRID	50
5.4	Table of figure.....	50

1 Introduction

1.1 Submitting Companies

- MEGA International
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1.2 Supporting organization

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1.3 Status of the Document

This document is the initial submission.

1.4 Guide to the Submission

- Section 2 describes the problem to be solved
- Section 3 discusses the design rationale.
- Section 4 specifies the metamodel
- Section 5 describes mappings of the metamodel to example languages
- Section 6 responds to the specific RFP requirements
- Section 7 specifies compliance points

1.5 Proof of Concept

These specifications are based on understanding of the metamodels of existing business modeling solutions, such as BMM and business process modeling languages such as BPMN, as well as market recognized frameworks such as the business canvas, the value proposition canvas and the BizBok.

1.6 Submission Contacts

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1.7 Change History

June 2018	Initial submission

2 Problem to be solved

This specification defines a set of metamodels for “business architecture”. The metamodel will lay out a foundation for strategic enterprise transformation by providing an overarching view of the enterprise - sometime called a “conceptual” view of the enterprise - and its evolution over time. The concept of “capability” is at the core of this overarching and planning view. The word “capability” is indeed a controversial term and does not have the same meaning depending on the audience. There are different kinds and sub-kinds of capabilities. A major distinction can be made between strategic capabilities - the capabilities exposed by an enterprise as part of its business model – and operating capabilities – the capabilities which define the functional structure of the enterprise. The neat distinction of these two aspects of capabilities provides a clarification of the relationships between capabilities and the other dimensions of the BACM RFP: strategy, product & services, value stream, projects, organizations. The introduction of a spatio-temporal view of the enterprise will provide a new perspective for enterprise transformation. Combined with the loosely coupled principle of the systemic approach, BACM is creating the foundation for an agile and responsive business architecture framework.

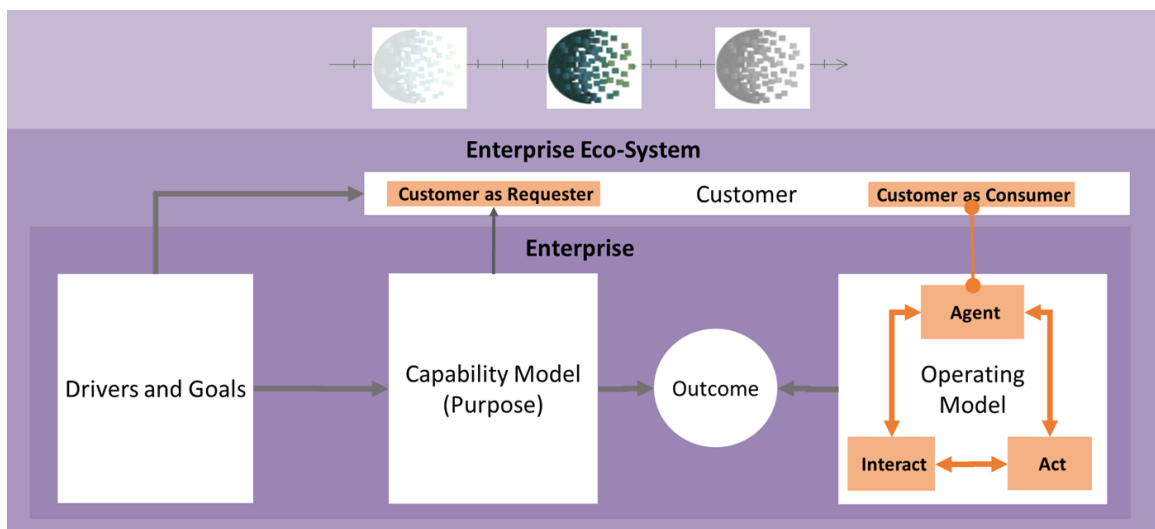


Figure 1 – BACM Scope: Motivation/Capability/Operating Model – All over time

All these aspects of business architecture share a common structure, also called upper-ontology which provide a semantic unification for the various meta-models. The value is not to deliver a universal metamodel for business architecture, but rather to provide an integration framework for each concrete metamodels ensuring that they can be related to one another.

The specification developed in response to this RFP is expected to achieve the following:

- Provide a Meta-Model for strategic capabilities and business models.
- Provide Meta-Model(s) to create both business models and business operating models as the motivations that drives the strategic directions of the enterprise over time (capability planning).
- Ensure that BACM is aligned with connected initiatives:
 - The proof that the proposed Meta-Model is suitable for BPMN types of processes.
 - The proof that the proposed Meta-Model is suitable for BMM strategy models.
 - The proof that the proposed Meta-Model is suitable for UAF models.
 - The verification that information models can be further described using UML class models.

The objective of BACM is more to take into account existing business modeling techniques than to ignore them: it is part of the BACM to align with BPMN, BMM, UAF, SBVR and UML. In addition, it is part of the BACM's charter to allow for enterprise transformation.

This introduction has the objective of explaining the core structure of the Meta-Model, and to give its rationale.

2.1 Building Block Pattern

Modern architecture descriptions need to address the ability to build architecture alternatives, to compare alternative architectures and be prepared for enterprise transformation while mastering enterprise complexity. However, the two-common modeling syntaxes used for architecture descriptions – monolithic hierarchies and flat models - prevent from defining building blocks with well-defined boundaries, thereby denying the very notion of building block.

The building block pattern aims at providing a structured syntax for enterprise modeling. The ability to defined clear boundaries for model elements deliver a syntaxial unit for the management of models as well as foundations for a modular approach of enterprise architecture. The building block and associated composition pattern follows the principle of “near-decomposability” described by Herbert Simon in his book “the science of artificial”:

On theoretical grounds we could expect complex systems to be hierarchies in a world in which complexity had to evolve from simplicity. In their dynamics hierarchies have a property, near decomposability, that greatly simplifies their behavior. Near decomposability also simplifies the description of a complex system and makes it easier to understand how the information needed for the development or reproduction of the system can be stored in reasonable compass.¹

In BACM, all concepts are kinds of “building blocks”. These are structurally composed through “building blocks components” and classified by their class of building blocks with the following benefits:

- Provision of catalogs of autonomous architecture building blocks. A building-block approach is the foundation for agile and value-added architecture endeavors.
 - Provision of alternative assemblies of these building blocks in response to enterprise planning.
 - Comparison of alternative assemblies.
 - Support for enterprise transformation though modular change impacts.
 - Support for multi-layered approaches to master enterprise complexity.
- See Herbert Simon: Parable of the two watchmakers.

¹ Simon, Herbert A.. *The Sciences of the Artificial* (MIT Press) (p. 216). The MIT Press. Kindle Edition

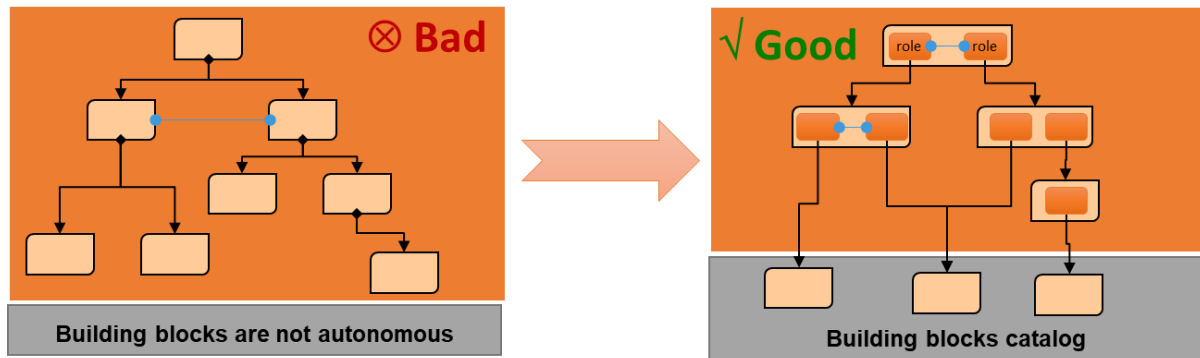


Figure 2 – Composite structures and modularity

2.2 Temporal aspects

One of the difficult challenge of BACM is to address enterprise transformation and thereby to introduce time in enterprise architecture descriptions. The approach followed by the BM-BOM submission is called the 4D approach which comes from ESPISTLE European research projects and has led, among others, to the ISO-15926 ²standardization. The IDEAS project, led by a consortium of ministries of defense, has extended the 4D approach from ISO 15926 to enterprise architecture. It is today at the foundation of the OMG UAF specification. The 4D principle is that the identity of objects is based on their spatio-temporal extent. This comes back to the story of the "knife of toto" which handle has been changed, followed by the change of its blade. The question arises whether this is still the same knife. In the 4D approach, each state of the knife is considered as an entity by itself, resulting in a chain of states expressing the knife history over time. It is this "history" that constitutes the identity of the object.

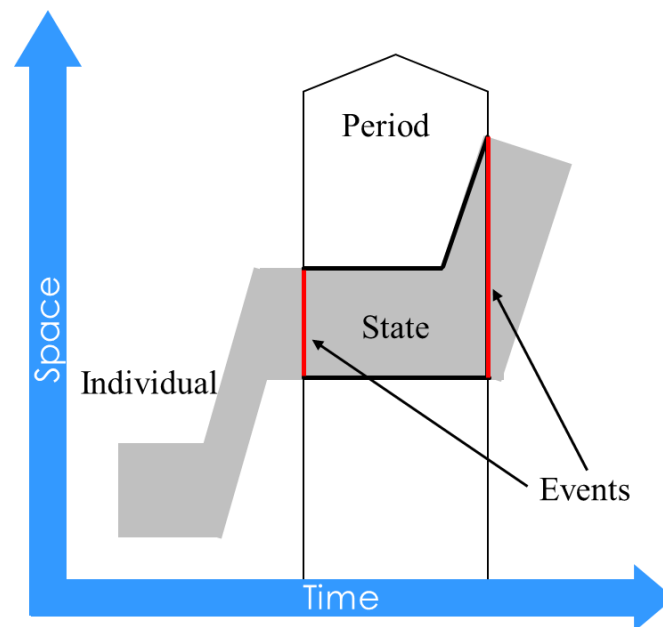


Figure 3 – 4D approach of identify over time

² Official site for ISO 15926: <https://www.iso.org/standard/29557.html>

Resources for navigating ISO 15926: <http://www.15926.org/topics/data-model/index.htm#diagram> for top level

Trivial in appearance, this question of identity - what is enduring and what changes over time - is a theme that has been pending from the origin of philosophy. In his book "Oneself as Another" Paul Ricoeur discusses this central topic and cites the examples on which Locke relies. They do not differ fundamentally from the knife history:

In the first series of examples - the ship of which all the pieces have been changed, the oak which accompanies the growth of the acorn to the tree, the animal and even the man whose development is followed from birth to dead - it is the truth that prevails; the element common to all these examples is the permanence of the organization, which, it is true, does not commit, according to Locke, any substantialism..³

While moving from the simplistic history of the knife to the philosophical questions of Locke and Paul Ricoeur, it might seem that the subject is getting outside BACM concerns. However, the 4D ontology provides a simple and elegant answer to these questions. Matthew West describes thereby the following mantra of 4D:

1. *Individuals exist in a manifold of 4 dimensions, three space and one time. So things in the past and future exist as well as things in the present.*
2. *The four dimensional extent is viewed from outside time rather than in the present.*
3. *Individuals (including physical objects) extend in time as well as space and have both temporal parts and spatial parts.*
4. *When two individuals have the same spatio-temporal extent they are the same thing. (However not all version of 4D insist on this principle).*
5. *Thus a 4D object is not (usually) wholly present at a point in time, but its whole is extended in space as well as time. The object at a point in time is a temporal part of the whole. Change is naturally expressed through a 4 dimensional classical mereology, which Simons: "Parts: A Study in Ontology" in his seminal work, describes in one page. A good description of, and argument for, the 4D paradigm can be found in Sider: "4 Dimensionalism".*

One of the key answers to the question of identity is locate in topic 4: "When two individuals have the same spatial-temporal extent, they are the same thing". This makes it clear what is meant by "individual". These are the things that have their own existence and whose history unfolds over time. It is also interesting to read the definition "Possible Individual" in ISO-15926. It provides details on what is called the "temporal parts": *A PossibleIndividual is a Thing that exists in space and time.*

- *This includes:*
 - *things where any of the space time dimensions are vanishingly small,*
 - *those that are either all space for any time, or all time and any space,*
 - *the entirety of all space time*
 - *things that actually exist, or have existed,*
 - *things that are fictional or conjectured and possibly exist in the past, present or future,*
 - *temporal parts (states) of other individuals,*
 - *things that have a specific position, but zero extent in one or more dimensions, such as points, lines, and surfaces.*

³ Ricoeur, Paul : [Oneself as Another](#)

- *In this context existence is based upon being imaginable within some consistent logic, including actual, hypothetical, planned, expected, or required individuals.*

EXAMPLE The pump with serial number ABC123, Battersea Power Station, Sir Joseph Whitworth, Shakespeare, and the starship "Enterprise" can be represented by instances of PossibleIndividual.

The choice made here on the "temporal parts" is decisive: changes on individuals are handled through a chain of states rather than a set of changes on relationships. Relationships, as well as classes are immutable. In the example of the knife, the relationships between the knife and its first handle will be kept in state 1. States 2 and 3 will have the same relationship to the new handle.

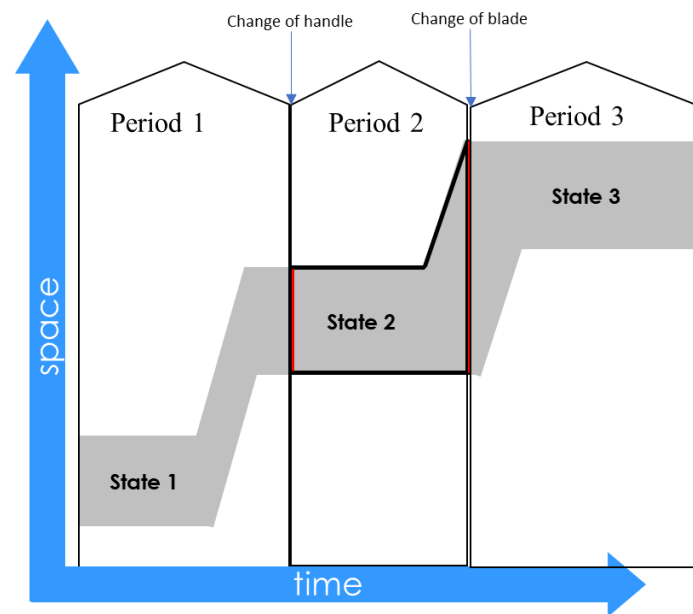


Figure 4 – Knife over time

2.3 Purposeful Systems and Capabilities

BACM is grounded in a polished systemic approach which provides enhancements and clarifications to the traditional system viewpoint. This traditional viewpoint has been summarized by Jean-Louis Le Moigne in his book "Theory of general system".

A system is:

- A thing (anything, that is assumed to be identified),
- which in something (environment)
- for something (finality or project)
- does something (activity = functioning),
- by something (structure = stable form),
- which transforms itself overtime (evolution).

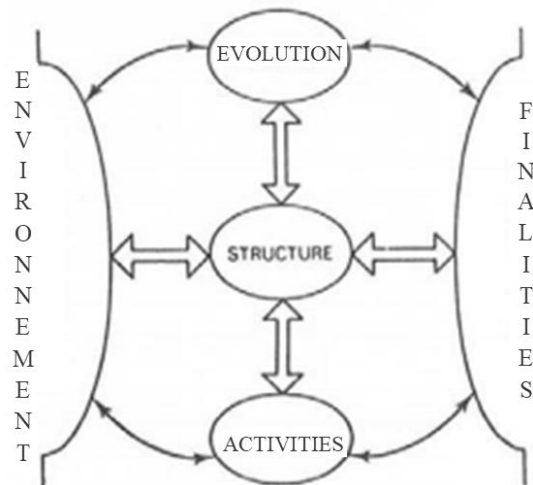


Figure 5 – Traditional systemic viewpoints

While most of system aspects are considered, the traditional approach fails in distinguishing finalities (purposes) from projects as well as operating a proper integrating between structures and activities.

2.3.1 Finality & capabilities: goals versus purposes

The finality issue is at the core of the BACM endeavor. Drivers and Goals, indeed, are about volition that comes as a reaction to the situation of the system in its environment. They shape and define wishes about the becoming of the system. Hence goals and drivers are always time bound. This becoming is then expressed in terms of what the system should be able to do: its exhibited capabilities. Finally, capabilities themselves are defined as *abilities to produces an outcome*. WordNet provides the following definition for ability:

Ability: the quality of being able to perform; a quality that permits or facilitates achievement or accomplishment⁴.

Capabilities are understood as “qualities”, which are emerging properties. They are not performers or agents and as such they have no “behavior”. A second aspect of capabilities is the *achievement* that they permit: their outcomes. Capabilities cannot be understood without their outcome. In the service-oriented view of systems adopted by BACM, capabilities represent the *offer* aspect system architectures, as stated in the OpenGroup white paper “Defining IT Services”.

An offer <..>. describes available capabilities from the provider in value proposition terms (outcome-oriented).⁵

⁴ WordNet 3.1: ability:

<http://wordnetweb.princeton.edu/perl/webwn?o2=&o0=1&o8=1&o1=1&o7=&o5=&o9=&o6=&o3=&o4=&s=ability&i=0&h=00#c>

⁵ OpenGroup : W161 page 10: <https://publications.opengroup.org/w161>

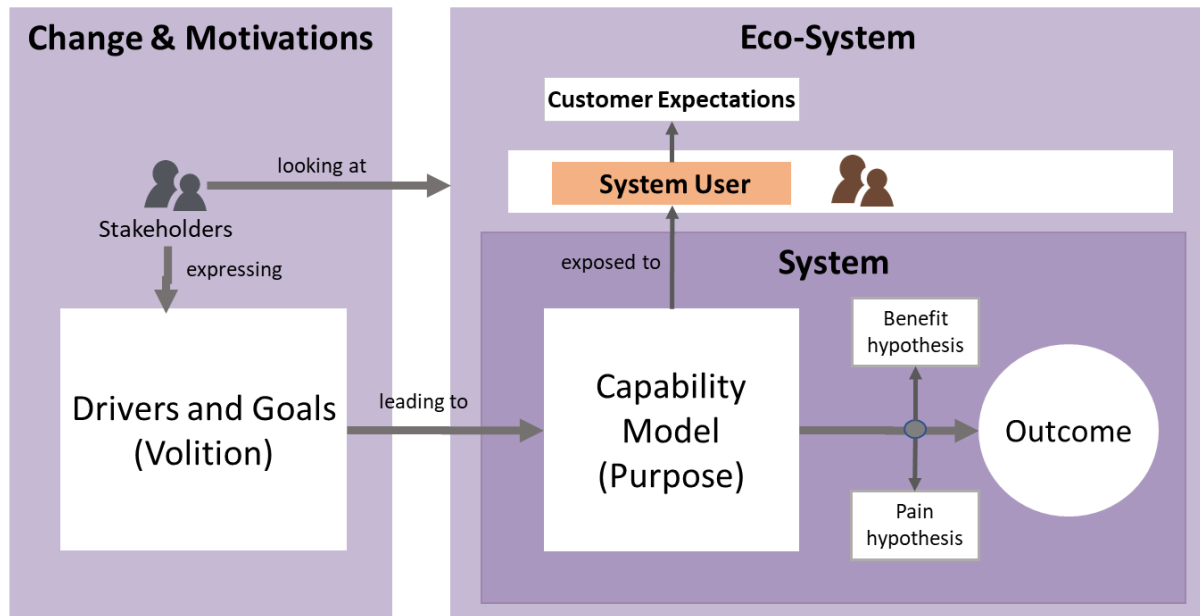


Figure 6 – Goals & Purposes (Capabilities)

The “capability + outcome” couple provides a formal solution to the product & service dilemma. The usual misunderstanding comes from considering that products and services are at the same level. This is obvious stated in sentences such as “enterprises deliver products and services”. In fact, one shall rather say services & products. Products are indeed, outcomes (results), while services are acts through which those outcomes are provided. Capabilities are the emerging properties of systems in terms of:

- Desired outcomes of systems.
- Ability and associated qualities leading the delivery of these outcomes.

Let’s take two examples: pizza delivery and haircut. In pizza delivery, the outcome is the delivered pizza. It has qualities such as “being crispy” and “being hot”. The delivery itself also has qualities: “being on time”, “being friendly”, “being available 24/24”. These qualities are on the ability to perform the delivery: the “pizza delivery” capability. In haircut, the outcome is the final dressed hair on a body. It has specific qualities such as: “hair style”, “hair color”, “hair brightness”. The hair cutting capability itself also has qualities: “customer care”, “waiting time”, “ambiance”.

Capabilities are the functional “raison d’être” of systems: their purposes.

2.3.2 The operating triptych: system, action, interaction

In front of offered capabilities comes a functioning system. It is the place where actions take place to deliver desired outcomes. Following the OpenGroup white paper, we shall notice that offered capabilities translate into interactions at the edge of the functioning system:

When described as a performance or an act, every service will have three key aspects: an interaction where a performance of the service occurs, an offer that describes available capabilities from the provider in value proposition terms (outcome-oriented), and a service system which is the integrated set of resources needed to facilitate the interaction/outcome.

<..>

The primary difference with this work is that we believe the separation of concerns between the act (interaction), capability (offers), and actors (service systems)⁶

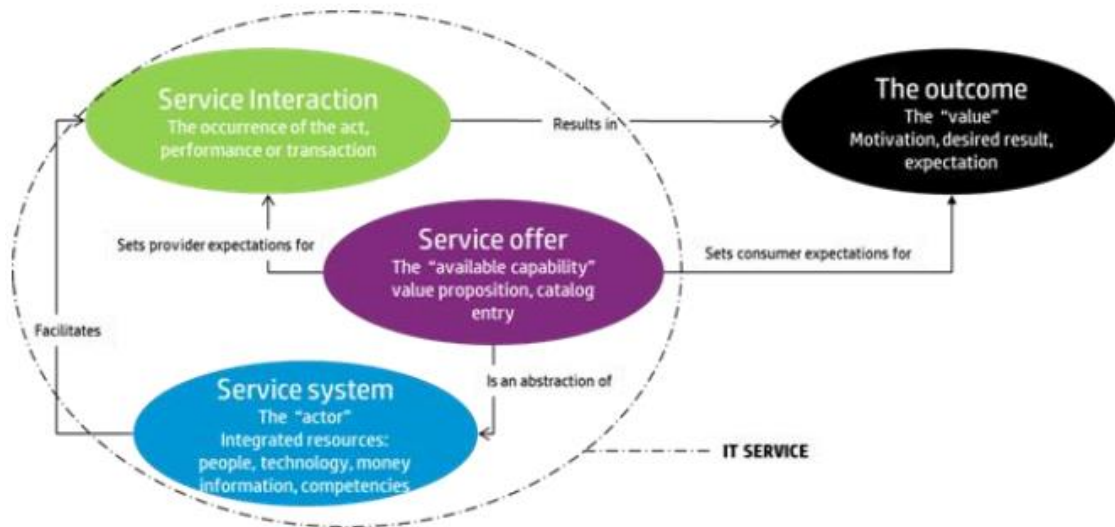


Figure 7 – IT Service in OpenGroup W161 whitepaper

This clarification is a key point to avoid the confusion that usually arises between service and system. A step beyond is to articulate the external structure of systems - where interactions occur - with their internal structure where actions are distributed among sub-systems to achieve the desired outcome. The art of system architecture is to balance the distribution of actions & interactions among sub-system, which form the operating triptych - system, action, interaction of any operating architecture.

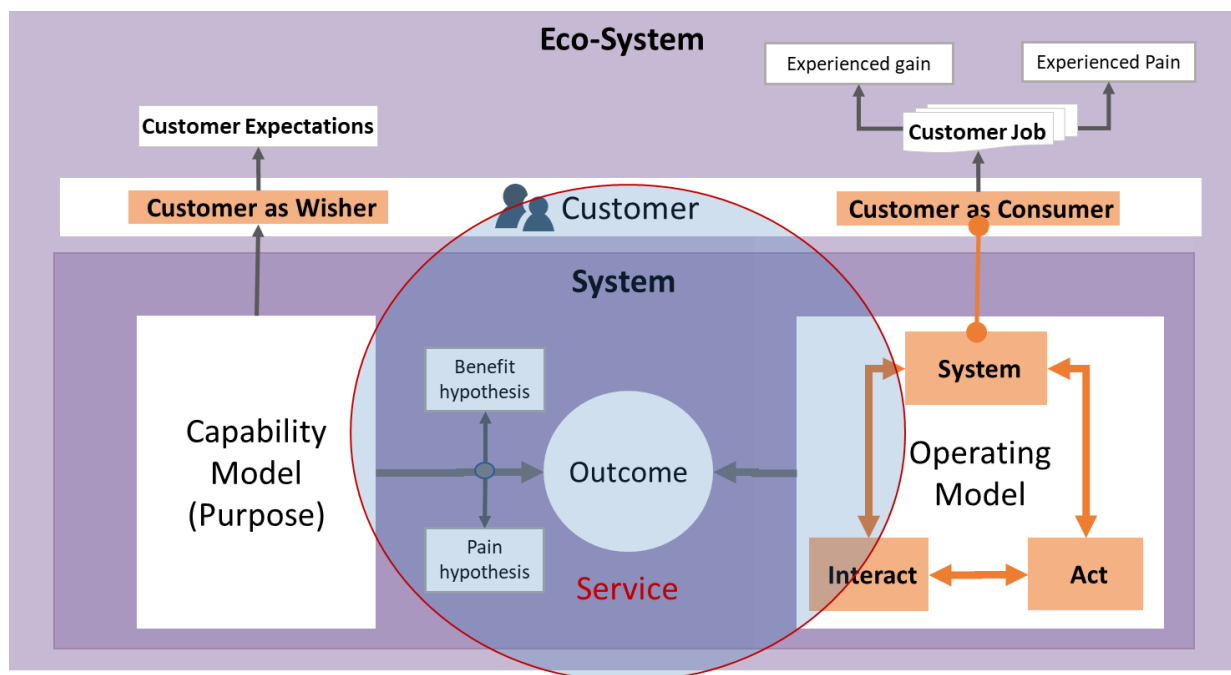


Figure 8 – The operating triptych and capabilities

As a summary, any system architecture shall be considered according to three axes:

⁶ OpenGroup : W161 page 12: <https://publications.opengroup.org/w161>

- The motivation for creating/changing/adapting the system: goals are different from purposes as they express a becoming.
- The purpose of a system: the functional raison d'être of a system expressed in terms of what it should be able to do: its capabilities and associated desired outcomes.
- The functional structure of a system: the operating triptych which expresses the way systems act and interact in order to deliver desired outcomes.

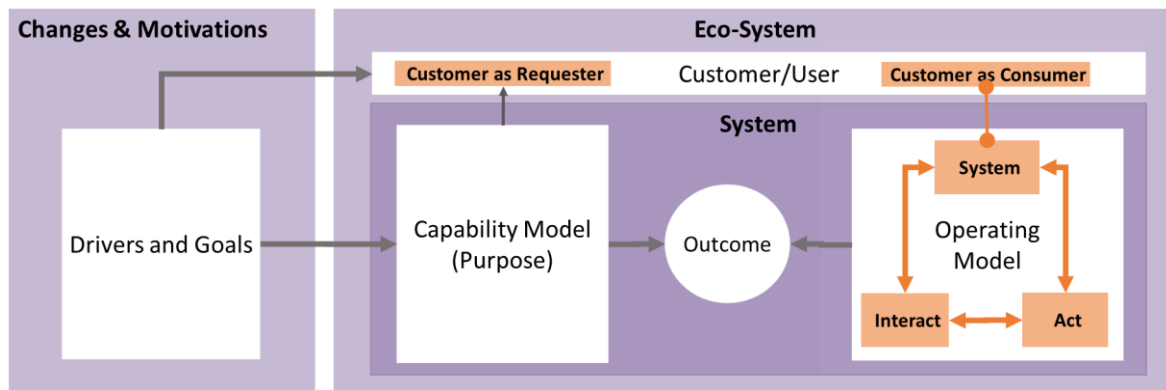


Figure 9 – System & Purpose Approach

2.4 From system to the enterprise

Enterprise are often considered as a system of system. They have the specific characteristics of any purposeful organization that Russell Ackoff has described below:

Type of System	Behavior of System	Outcome of Behavior
State-Maintaining	Variable but determined (reactive)	Fixed
Goal-Seeking	Variable and chosen (responsive)	Fixed
Multi-Goal-Seeking and Purposive	Variable and chosen	Variable but determined
Purposeful	Variable and chosen	Variable and chosen

Figure 10 - Russell Ackoff classification of systems

An organization is a purposeful system that contains at least two purposeful sub-systems which have a common purpose. An organization has a functional division of labor in pursuit of the common purpose(s) of its elements that define it. The functionally distinct sub-systems can respond to each other's behavior through observation or communication.⁷

⁷ Russell Ackoff: Towards a System of Systems Concepts:

https://www.researchgate.net/publication/227443596_Towards_a_System_of_Systems_Concepts

As purposeful systems, Enterprises define their own motives to change. Their ability to evolve depends both on the way they construction an understanding of heir eco-systems and on their ability to adapt their operating model.

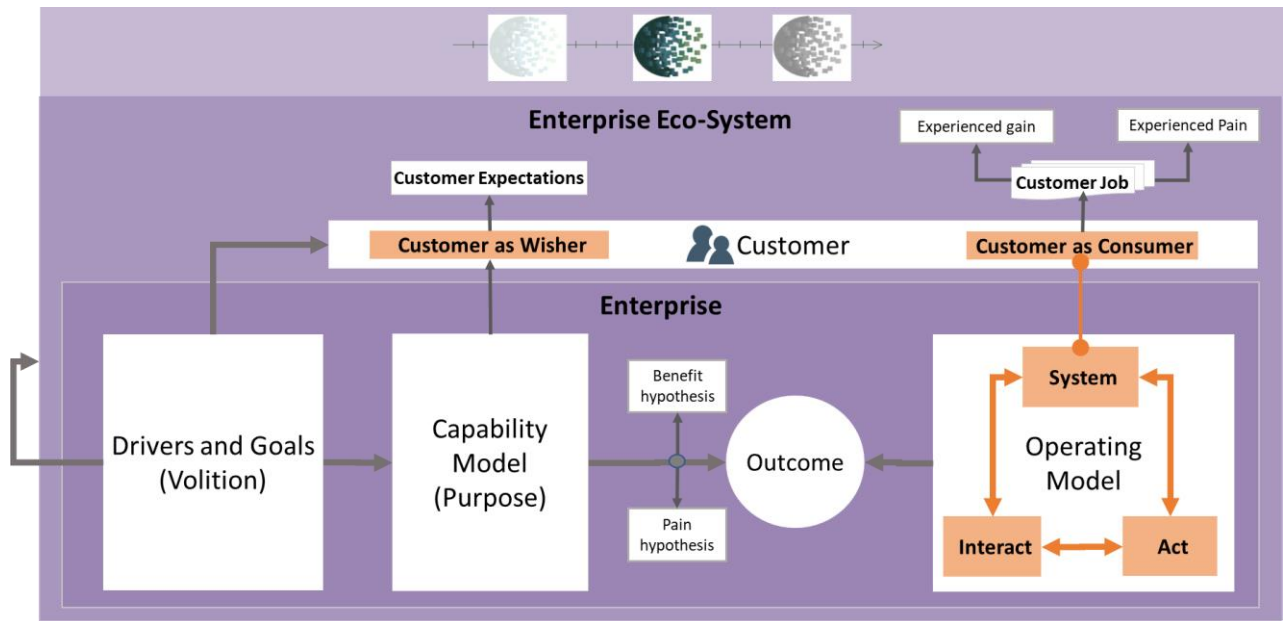


Figure 11 – The Enterprise as an evolving system defining its own purposes overtime

2.5 Gap analysis and Transformation

Gap analysis is sometimes used as a synonym for alignment which covers three different aspects:

- Architecture Alignment: filling the gaps between solutions architectures and reference architectures. For instance, an alignment between a physical data model and a conceptual data model is an architecture alignment.
- Resource usage Alignment: filling the functional gaps between functionalities required by human operator to achieve their tasks, and the artifacts they use to fulfill these needs.
- Transformation: filling the gaps between as-is and to-be architectures.

BACM addresses these three aspects of transformation with a scope limited to organization alignment for “architecture alignments”.

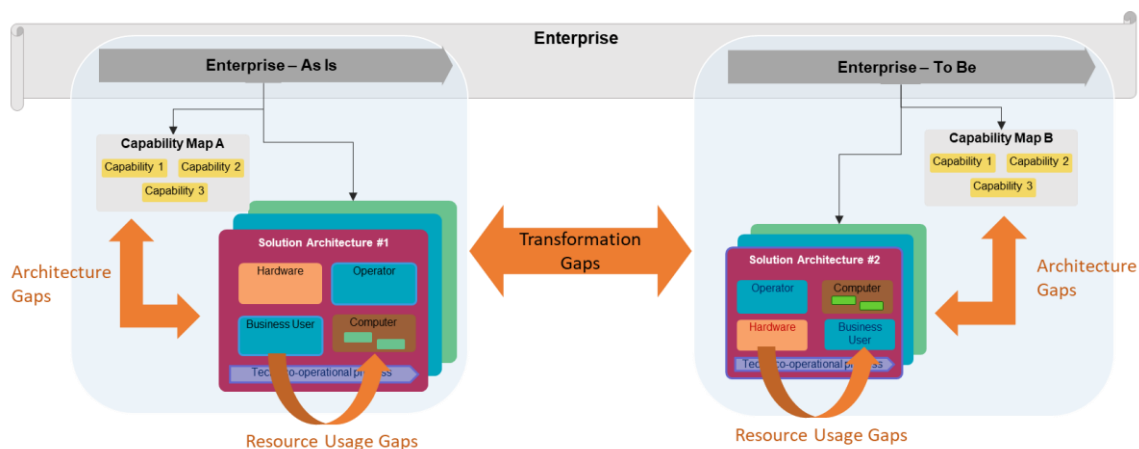
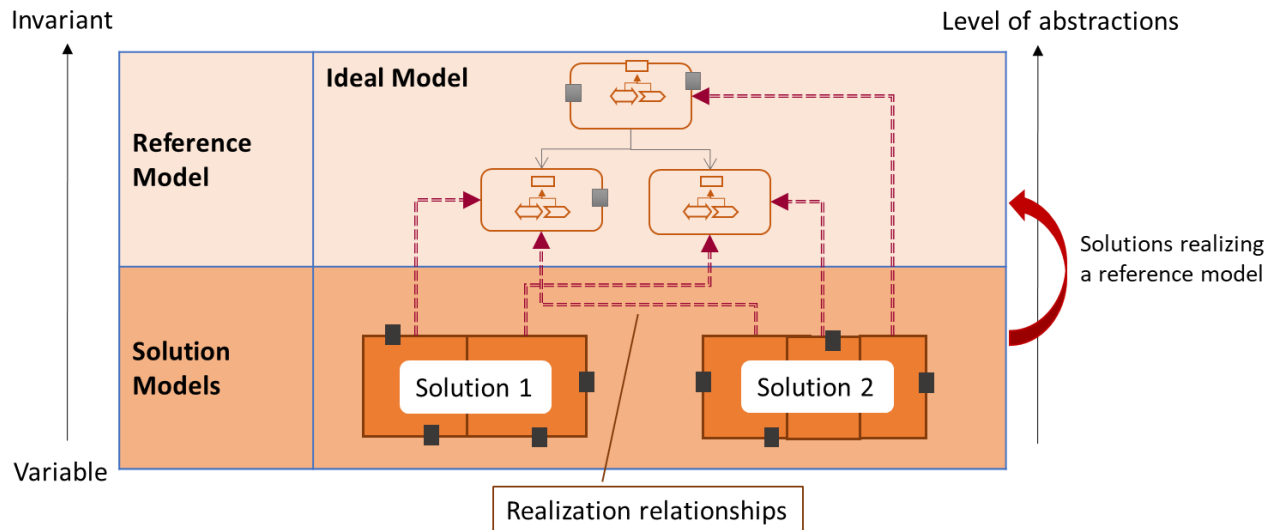


Figure 12 - Various kinds of gaps and corresponding alignments

Any architecture alignment implies a gap analysis between a reference model and one or more solutions realizing this reference model. The alignment approach consists in analyzing these gaps through realization relationships that express how solutions implement the reference model.

Reference models are always more “abstract” than solutions models.



Operating Models define how the enterprise operates. They can be defined at a conceptual level – the Business Operating Model – or at a solution level -Solution operating models.

Business Operating Alignment aims at providing the best fits between Solution Operating Models to their reference Business Operating Model.

2.6 BACM concept GRID

BACM provide a grid to be able to navigate among its various concept. Columns represent the various aspects of enterprise architecture: capability, outcome, agent, process, etc. Rows represent the layers of abstractions under which enterprise architecture is considered: conceptual business layers, human resource layers, other physical resources layers.

	Purpose Analysis Models		Operating Models					Qualification Models	
Object Kind →	What is expected	What is produced	How outcomes are delivered	Who is in charge	How things are done	What shall be remembered	How things are ruled	How to quantify	How things are classified
↓ EA Layer	Capability	Outcome	Services	Agent	Process	Information	Policies	Measures	Categories
Business Layer	🧑 Business Architect, Strategic Planner								
Organization Layer	🧑 Business Analyst								
Deployed Resources	🧑 Other resources								
Motivation & Transformation	🧑 Enterprise Architect, Strategic Planner								

Figure 13 - BACM simplified GRID

3 BACM & Existing Standards

3.1 BACM and UML 2.0 & MOF

UML 2.0 provides a powerful set of infrastructure artifacts -class, package, feature - to build the backbone of a business architecture framework. UML 2.0 concepts are reuse wherever they match the corresponding BACM business concepts. MOF based concepts are created where UML 2.0 equivalent concepts do not exist or do not fit to common business semantic.

3.2 BACM and BPMN

BPMN is intended to be the process model that comes along with BACM. BACM is compatible with the adopted BPMN 2.0 version and provides an extension to BPMN to fulfill the requirements of BACM concepts. The version of BPMN aligned with BACM is BPMN 2.0.

3.3 BACM and BMM

The core concepts of the BMM provide the foundation for strategy modeling in BACM. The concepts N is intended to be the notation that comes along with BACM. BACM is compatible with the adopted BPMN 2.0 version and provides an extension to BPMN to fulfill the requirements of BACM concepts. The version of BPMN aligned with BACM is BPMN 2.0.

3.4 BACM and UAF

The intent of BACM is to align with the global Enterprise Architecture framework provided by UAF. UAF comprise both a domain Meta-Model and a UML profile. This specification is referring to the UAF domain Meta-Model.

3.5 BACM and Business Model Canvas

The business motivation canvas is a well establish tool a present in a synthetic manor the various aspect of an enterprise business model.

BACM aims at providing a formalization of the Business Model Canvas with two objectives in mind:

1. Checking that BACM concepts match to establish business practices.
2. Providing powerful analytic tooling to the business canvas by leveraging the power model driven business architecture

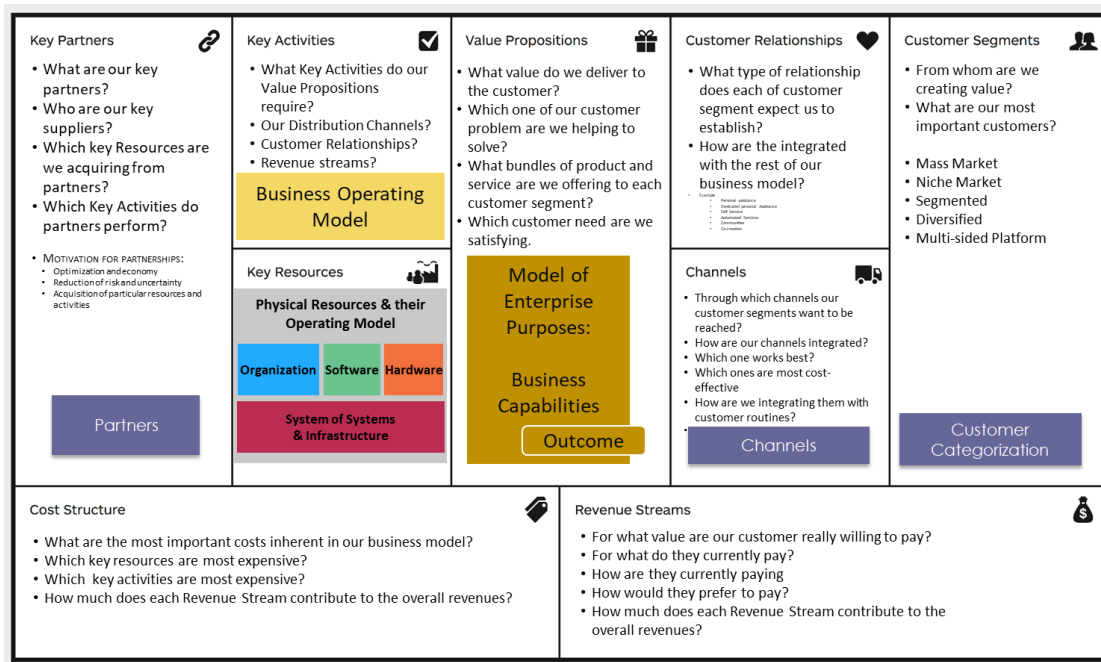


Figure 14 6 – Business Model Canvas

3.6 BACM and Value Proposition Canvas

BACM use the Value Proposition Canvas as a framing tool for value description. BACM will provide a formal representation of the Canvas for value analysis.

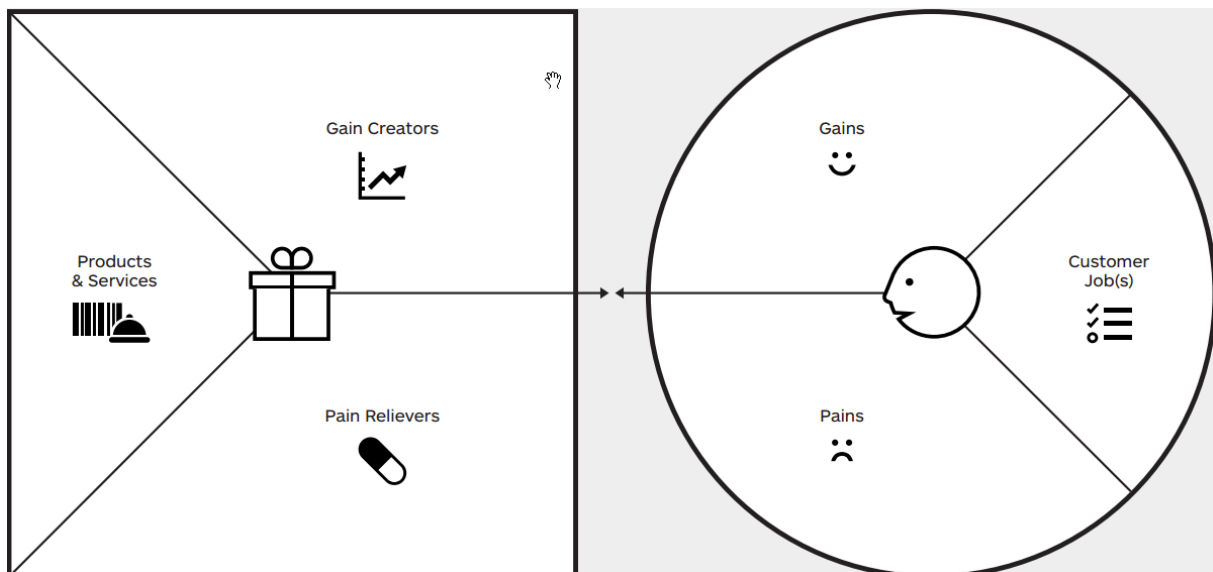


Figure 15 – Value Proposition Canvas

4 Specification

This section presents the normative specification for business architecture core metamodel. It begins with an overview of the BACM metamodel structure followed by a description of each sub-package.

4.1 Overview

BACM holds all definitions common to business models and business operating models. It also establishes relationships between processes, information and strategy models.

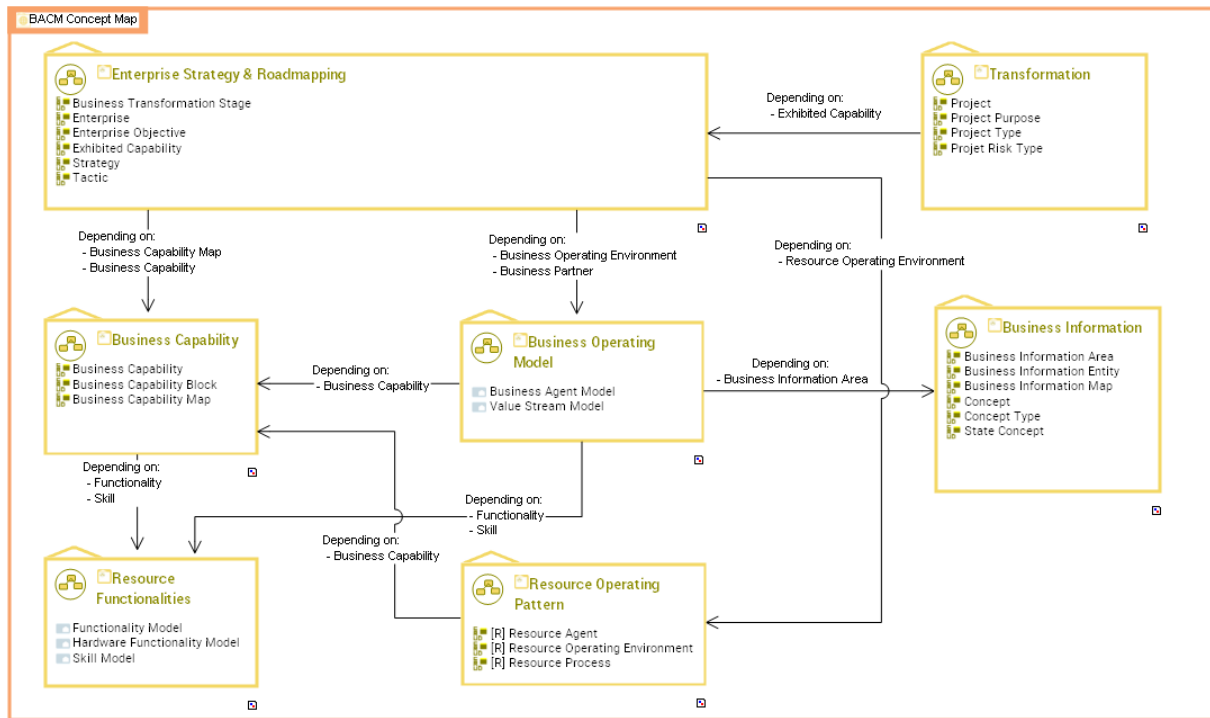
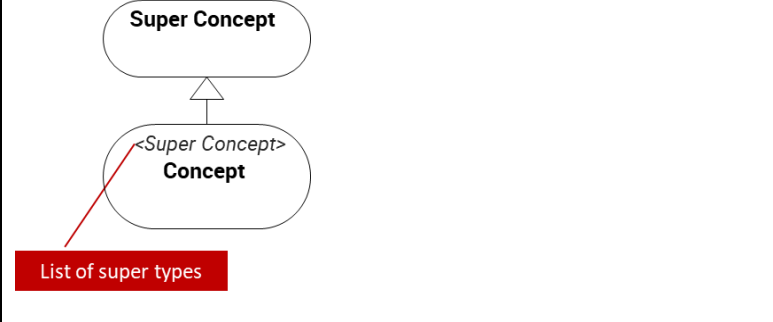
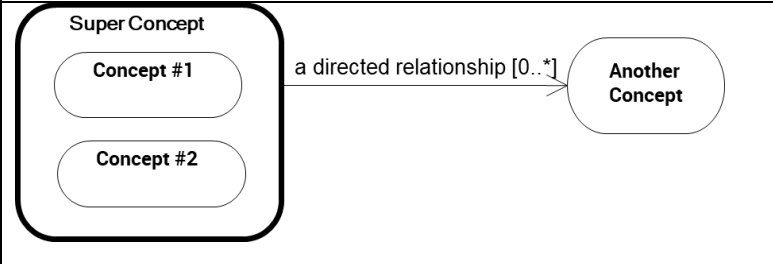


Figure 16 – BACM Concept Map

Strategy Package	Business Motivation Objects
Business Capability	Definition of strategy capability maps and their relationships to resource capabilities.
Business Outcome	Definition of products in relationship to business capabilities.
Business Operating Model	The operating triptych which expresses the way agents operate to deliver desired outcomes.
Resource Functionalities	Capabilities of physical resources that are required from the business operating model.
Transformation	The projects and transformation runway required for continuous architecture supporting enterprise transformations.
Business Information	The business concepts of the enterprise domain.
Resource Operating Pattern	The hook for physical operating models implementing the business operating model.

4.2 Notation convention

The BM-BOM submission uses an extension of the UML Class notation in order to offer a compact representation of sub-typing and to handle the differences between class and class of classes

	Any concept whether it is an individual or a class of individual. The list of super-types can be displayed at the top in order to provide a more compact notation
	Alternative Box in Box notation for super-types, used to highlight inherited relationships for a group of direct specialized concepts.
	Syntax model element which define the structure of
	Directed relationships.
	Individual building block
	Class of building block

4.3 Upper Ontology & EA Patterns

As any language, modeling language have three aspects:

- Syntax is the required grammar and punctuation of the language: The top-level ontology is about syntax.
- Semantics is about meaning - what do we mean by a Capability? The BACM GRID is about semantic in Enterprise Architecture.
- Pragmatics has to do with how to use models (modeling technic). What kind of model to use to address stakeholder concerns (method)? Example: how shall one use capability modeling in enterprise transformation projects?

The upper ontology comprises a syntax layer where the concepts of objects, relationships and building blocks are define, an individual/class layer and an architecture layer where the foundation for all Enterprise Blocks is setup.

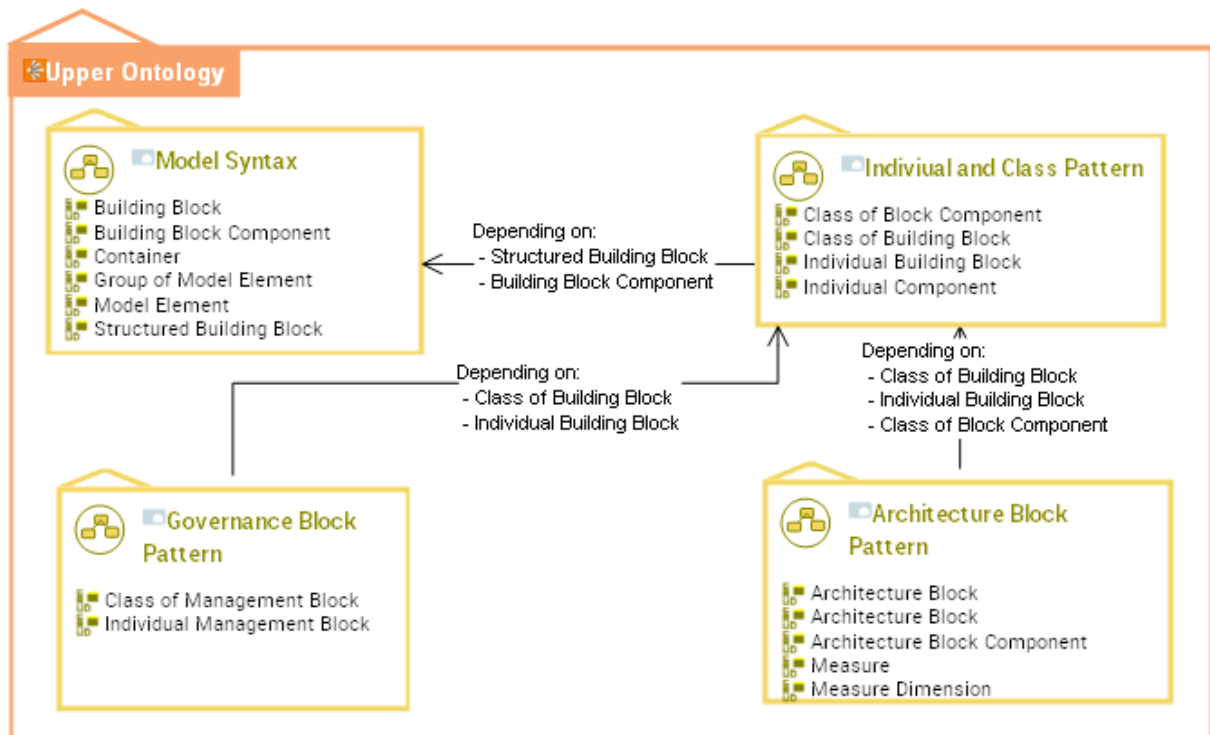


Figure 17 6 –BACM Concept Map

4.3.1 Building Block Pattern

The building block pattern contains root elements of the ontology where the syntax for all model is defined. It includes a two-level syntax: model elements and their grouping, building blocks and their composition.

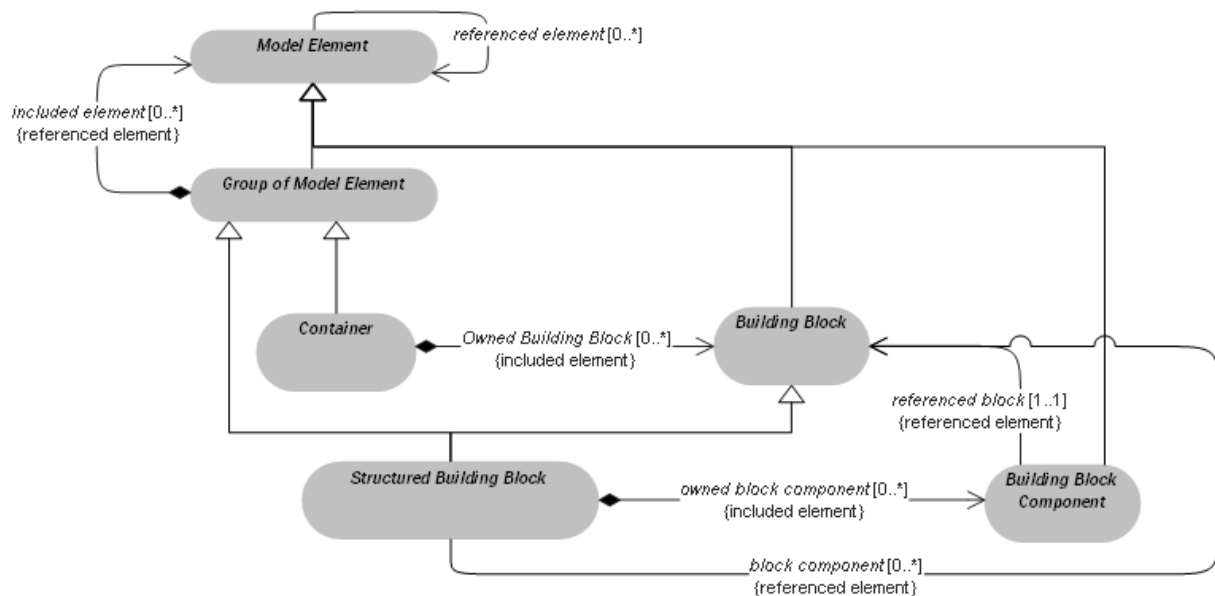


Figure 18 6 – Syntax Meta-Model

Building Block Pattern	
Building Block	Concept 1. A single undivided Model Element occurring in the composition of other Structured Building Blocks. Building Blocks are defined in the context of a Container which play

	the role of universe of discourse. (Hypernyms) Model Element (Hyponyms) Structured Building Block
Building Block Component	 Concept 1. A Model Element which represents a structured composition of a Building Block within a Structured Building Block. (Hypernyms) Model Element (Hyponyms) Class of Block Component, Individual Component (Component 1) referenced block: Reference relationship that (Type) Building Block (Presence) Always (Cardinality) 1 (Hypernyms) referenced element (Hyponyms) block in role
Container	 Concept 1. A Group of Model Element which plays the role of universe of discourse for the Building Block The set of building-blocks included in Container are asserted to exist for a given understanding of a business domain by a semantic community. Building Blocks of a universe-of-discourse are said to be declared by their universe-of-discourse which plays a role of package for its building-blocks. Example: Package in UML are containers. (Hypernyms) Group of Model Element (Hyponyms) Enterprise (Component 1) Owned Building Block: Inclusion relationship that asserts the existence of a given Building Block in the context of a given Container. A Building Block. belongs to one and one one Container. (Type) Building Block (Presence) Optional (Cardinality) * (Hypernyms) included element
Group of Model Element	 Concept 1. A Model Element that can include other Model Elements. (Hypernyms) Model Element (Hyponyms) Container, Structured Building Block (Component 1) included element: A directed relationship that indicates that the source Model Element includes its target Model Elements. (Type) Model Element (Presence) Optional (Cardinality) * (Hypernyms) referenced element (Hyponyms) Owned Building Block, owned block component
Model Element	 Concept 1. An Element is a constituent of a model. (Hyponyms) Building Block, Group of Model Element, Building Block Component (Component 1) referenced element: A directed relationship between Model Elements. (Type) Model Element (Presence) Optional (Cardinality) * (Hyponyms) included element, referenced block
Structured Building Block	 Concept 1. A Building Block which is composed of other Building Blocks. (Hypernyms) Building Block, Group of Model Element (Hyponyms) Class of Building Block, Individual Building Block (Component 1) /block component: A single undivided Model Element occurring in the composition of other Structured Building Blocks. Building Blocks are define in the context of a Container which play the role of universe of discourse. (Type) Building Block (Presence) Optional (Cardinality) * (Hyponyms) architecture component

	(Component 2) owned block component: Inclusion relationship that indicate that a Building Block Component is part of a Structured Building Block. (Type) Building Block Component (Presence) Optional (Cardinality) * (Hypernyms) included element (Hyponyms) owned human resource participant, owned Class of Component, owned component
--	---

4.3.2 Individual / Class Pattern

The Individual/Class pattern expresses the foundational ontological distinction between individual blocks and class of block.



Figure 19 6 – Individual / Class Meta-Model

Individual / Class Pattern	
Class of Block Component	Concept Type 1. Building Block Component which represents the composition relationship of a Class of Building Block within another Class of Building Block. (Hypernyms) Building Block Component (Hyponyms) Architecture Block Component (Component 1) block in role: The Powertype of Individual Building Block. Examples: * Cars * Boats * Mountains * Capabilities * Organisations (Type) Class of Building Block (Presence) Always (Cardinality) 1 (Hypernyms) referenced block (Hyponyms) architecture block role
Class of Building Block	Concept Type 1. The Powertype of Individual Building Block. Examples: * Cars * Boats * Mountains * Capabilities * Organisations (Hypernyms) Structured Building Block (Hyponyms) Measure Dimension, Measure, Architecture Block (Component 1) owned Class of Component: Building Block Component which represents the composition relationship of a Class of Building Block within another Class of Building Block. (Type) Class of Block Component (Presence) Optional

	(Cardinality) * (Hypernyms) owned block component (Hyponyms) owned architecture component
Individual Building Block	Concept 1. Individual concepts represent objects of the real world such as, the OMG in 2017, the BACM RFP in 2018 or SAP R3 deployed in Argentina since 2015. Individual objects shall be made distinct from IClass of Building Block which they are instance of. (Hypernyms) Structured Building Block (Hyponyms) Individual Management Block, Individual Architecture Block, Architecture Block (Concept Type) Class of Building Block (Component 1) owned component: A Structured Building Block which represents a composition relationship of an Individual Building Block within another Individual Building Block. (Type) Individual Component (Presence) Optional (Cardinality) * (Hypernyms) owned block component
Individual Component	Concept 1. A Structured Building Block which represents a composition relationship of an Individual Building Block within another Individual Building Block. (Hypernyms) Building Block Component (Component 1) block in role: Individual concepts represent objects of the real world such as, the OMG in 2017, the BACM RFP in 2018 or SAP R3 deployed in Argentina since 2015. Individual objects shall be made distinct from IClass of Building Block which they are instance of. (Type) Individual Building Block (Presence) Always (Cardinality) 1 (Hypernyms) referenced block

4.3.3 Architecture Block Pattern & Measures

Architecture Blocks are concepts used to describe the architecture enterprise. They include capabilities, outcomes, agents, services, processes, information and policies. They shall be distinguished from “Governance Blocks” which are used to build and

Architecture Blocks can be composed of other Architecture Blocks. All Architecture Blocks can be classified by measures. Measure are themselves class of building blocks.

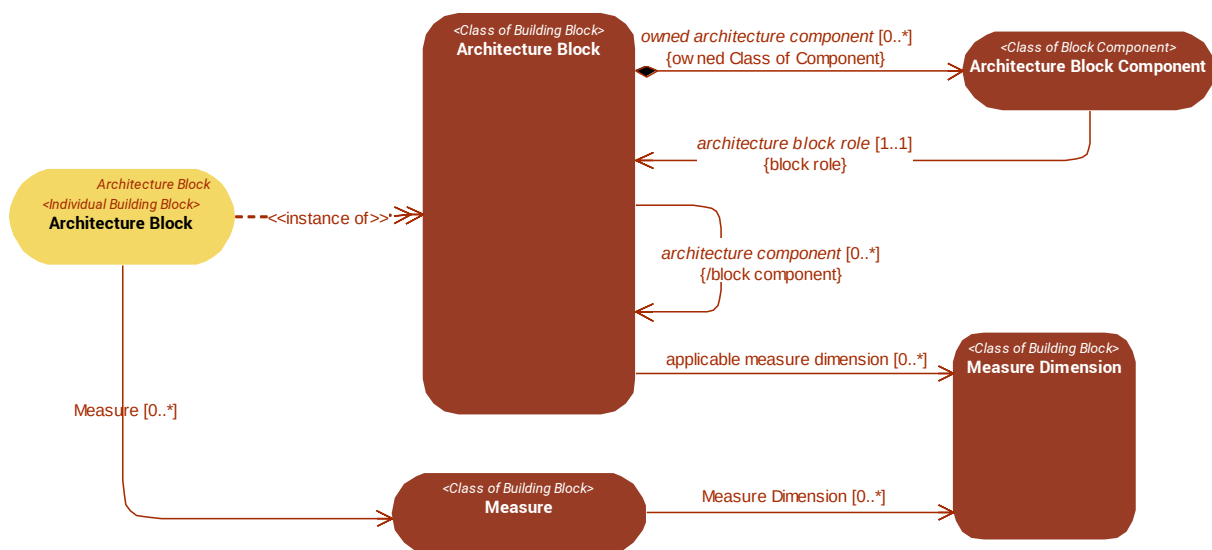


Figure 20 6 – Architecture Block Pattern

BACM - Temp	
Architecture Block	 Concept Type 1. A Class of Building Block which is used in the description of the enterprise, such as Process, Agent, etc. (Hypernyms) Class of Building Block (Hyponyms) Information Entity, Group of Information Area, Agent, Capability Building Block, Outcome, Process, Agent Operating Environment, Exchange Contract (Component 1) applicable measure dimension: Set of measure dimensions that can be applied to a Architecture Block . (Type) Measure Dimension (Presence) Optional (Cardinality) * (Component 2) owned architecture component: Class of Block Component which represents the composition relationship of a Class of Building Block] within another Architecture Block. . (Type) Architecture Block Component (Presence) Optional (Cardinality) * (Hypernyms) owned Class of Component (Component 3) architecture component: A Class of Building Block which is used in the description of the enterprise, such as Process, Agent, etc. (Type) Architecture Block (Presence) Optional (Cardinality) * (Hypernyms) block component (Hyponyms) consumed enterprise outcome, component information area, external Information Area, information entity component, entity in scope, desired outcome, fulfilled capability, exposed service, performed process, produced enterprise outcome, external capability component, capability component, Org-Unit Component, agent component, external agent component, subject resource agent component, information store  Concept 1. An Individual Building Block which is used in the description of the enterprise. Architecture Block can be classified by Measure (Hypernyms) Individual Building Block (Concept Type) Architecture Block (Component 1) Measure: The classification Architecture Block by Measures. (Type) Measure (Presence) Optional (Cardinality) *
Architecture Block Component	 Concept Type 1. Class of Block Component which represents the composition relationship of a Class of Building Block] within another Architecture Block. . (Hypernyms) Class of Block Component (Component 1) architecture block role: A Class of Building Block which is used in the description of the enterprise, such as Process, Agent, etc. (Type) Architecture Block (Presence) Always (Cardinality) 1 (Hypernyms) block in role
Measure	 Concept Type 1. How much there is or how many there are of something that you can quantify according to the considered Measure Dimension . Example : 30 minutes pizza delivery time, 20 kg, 1000\$. (Hypernyms) Class of Building Block (Component 1) Measure Dimension: The classification of a Measure

	by a Measure Dimension. . (Type) Measure Dimension (Presence) Optional (Cardinality) * (Component 2) Measure Dimension: The dimension the measure is an instance of. (Type) Measure Dimension (Presence) Always (Cardinality) 1
Measure Dimension	Concept Type 1. Measure Dimension Measure Dimensions express the nature of Measures (Time, Mass, Cost, ..) and define the units used for Measures (minutes, kg, euros, etc.). Example: - Time to deliver in minutes. - Delivery cost in kilo Euros. Measure Dimensions can be elementary or composite. Elementary dimensions are described by KPI units: kg, Liter, Gallon, Hour, Minute. (Hypernyms) Class of Building Block

4.3.4 Capability Pattern

Capabilities are understood as abilities to produce an outcome. Capabilities are properties which are fulfilled by agents. Depending on the considered EA architecture layers, there a different kind of capabilities to be considered. Hence, business capabilities address the business layer, skills address the human resource layer and functionalities address the various artifact layers.

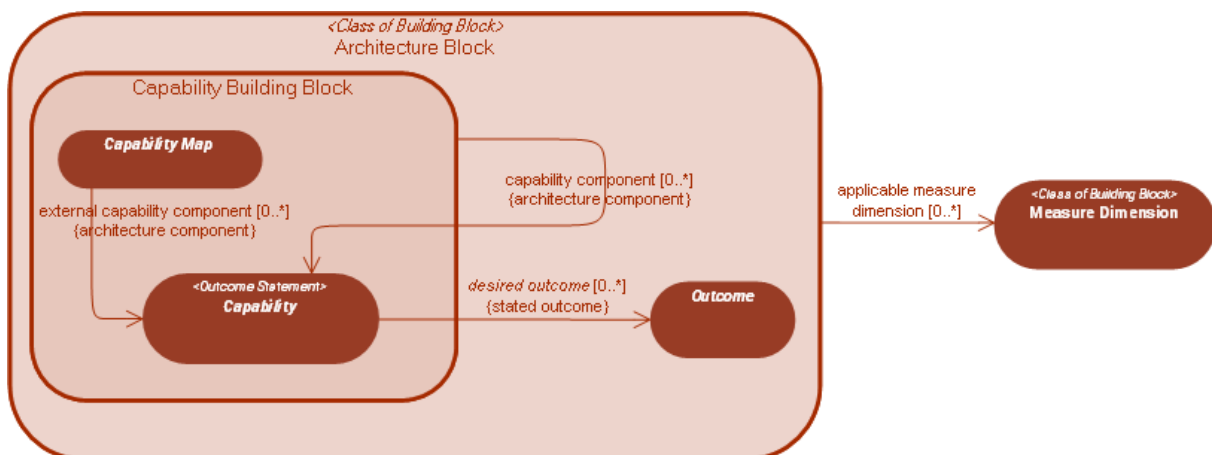


Figure 21 6 – Capability Pattern

BACM - Temp	
Capability	Concept Type 1. Ability to deliver an Outcome . Capabilities are fulfilled by Agents. (Hypernyms) Capability Building Block (Hyponyms) Functionality, Skill, Business Capability, Hardware Functionality (Component 1) desired outcome: Result of a process conducted by an Agent . (Type) Outcome (Presence) Optional (Cardinality) * (Hypernyms) architecture component (Hyponyms) desired resource outcome, desired

	hardware outcome, desired technical outcome, desired business outcome
Capability Building Block	Concept Type 1. Assembly of Capability and their dependencies. A Capability Building Block is either a Capability Map or a Capability. (Hypernyms) Architecture Block (Hyponyms) Capability Map, Capability, Business Capability Block (Component 1) capability component: A directed reference relationship between Capability Building Blocks and their Capabilities. (Type) Capability (Presence) Optional (Cardinality) * (Hypernyms) architecture component (Hyponyms) business capability component, functionality component, skill component, Hardware Functionality Component
Capability Map	Concept Type 1. Top level assembly of Capability and their dependencies that, together, provide a business model scope for Enterprise Phases. (Hypernyms) Capability Building Block (Hyponyms) Functionality Map, Business Capability Map, Skill Map, Technical Functionality Map, Hardware Functionality Map (Component 1) external capability component: Capability with is used an an external component of a Capability Map . (Type) Capability (Presence) Optional (Cardinality) * (Hypernyms) architecture component (Hyponyms) business capability external component

4.3.5 Process Pattern

The process pattern provides a general hook for the various kind of processes available in BACM. All detailed process artifacts, such as sequences, flows, events are provided by BPMN.

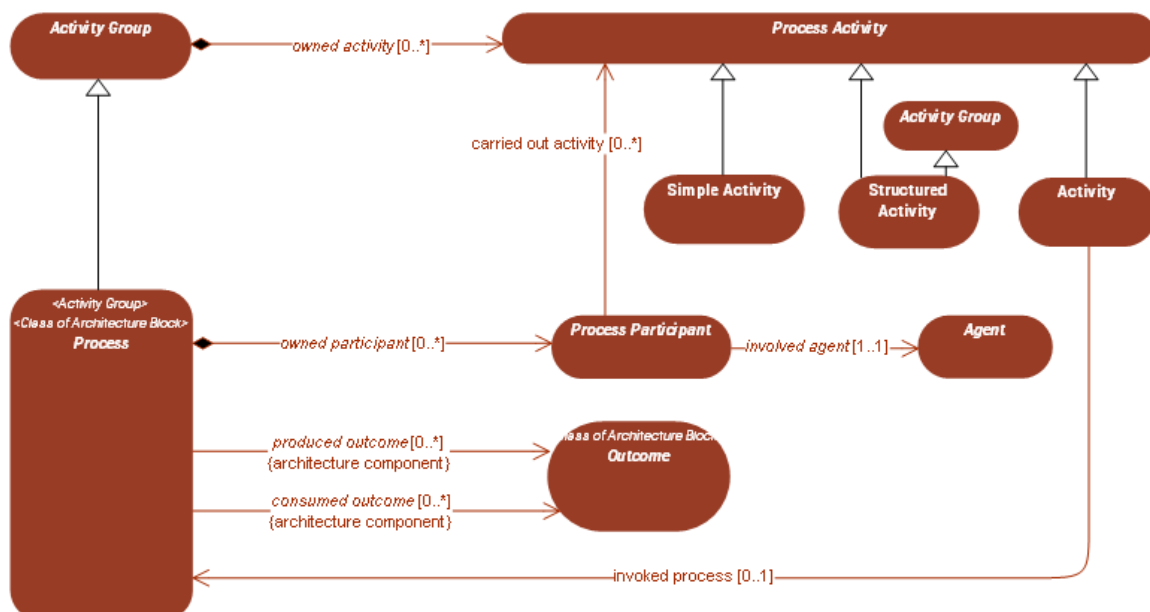


Figure 22 6 – Process Pattern

Process Pattern	
Activity	Concept Type 1. Unit of work which is an invocation of another Process.

	(Hypernyms) Process Activity, Architecture Block Component (Hyponyms) Resource Activity (Component 1) invoked process: Reference relationship that states which Process is invoked as part of an Activity. (Type) Process (Presence) Optional (Cardinality) 1 (Hypernyms) architecture block role (Hyponyms) invoked resource process
Activity Group	 Concept Type 1. A group of Process Activity . (Hypernyms) Group of Model Element (Hyponyms) Resource Activity Group, Structured Activity, Process (Component 1) owned activity: An inclusion relationships that stated that an Activity Group is made of Process Activity(ies). (Type) Process Activity (Presence) Optional (Cardinality) * (Hyponyms) owned value stage, owned resource activity
Process	 Concept Type 1. A typical course of action intended to achieve an Outcome through the involvement of Agents . (Hypernyms) Activity Group, Architecture Block (Hyponyms) Value Stream, Resource Process (Component 1) produced enterprise outcome: Outcome which is a result of the completion of a Process. (Type) Outcome (Presence) Optional (Cardinality) * (Hypernyms) architecture component (Hyponyms) produced enterprise outcome, produced resource outcome (Component 2) consumed enterprise outcome: Outcome which is required as an input of a Process. (Type) Outcome (Presence) Optional (Cardinality) * (Hypernyms) architecture component (Hyponyms) consumed resource outcome (Component 3) owned participant: Inclusion relationship that states that a Process is structurally composed of the involvement of Agents as Process Participants. (Type) Process Participant (Presence) Optional (Cardinality) * (Hypernyms) owned architecture component (Hyponyms) owned resource process participant, owned value stage
Process Activity	 Concept Type 1. Any unit of work done by an Agent involved as a Process Participant (Hypernyms) Model Element (Hyponyms) Simple Activity, Resource Process Activity, Value Stream Stage, Structured Activity, Activity (Component 1) Risk (Type) Risk (Presence) Optional (Cardinality) *
Process Participant	 Concept Type 1. Involvement of an Agent in a Process through which the Agent is assigned activities. (Hypernyms) Architecture Block Component (Hyponyms) Resource Process Participant, Value Stream Participant (Component 1) involved agent: Reference relationship that states which Agent is involved as a Process Participant. (Type) Agent (Presence) Always (Cardinality) 1

	(Hypernyms) architecture block role (Hyponyms) Involved Business Agent, involved resource agent (Component 2) carried out activity: A reference relationship that states that a Process Participant carries out Process Activity(ies). (Type) Process Activity (Presence) Optional (Cardinality) * (Hyponyms) carried out stage, carried out resource activity
Simple Activity	Concept Type 1. Simple, indivisible unit of work done by an Agent . (Hypernyms) Process Activity (Hyponyms) Resource Simple Activity
Structured Activity	Concept Type 1. Hierarchical grouping of multiple units of work done by an Agent . (Hypernyms) Process Activity, Activity Group

4.3.6 Operating Model Pattern

The operating model pattern is a triptych which expresses the way agents operate to deliver desired outcomes:

- Agents act: processes.
- Agent interact: service exchanges.
- Agent store: memorization and storage of information. There are no flows without stores.

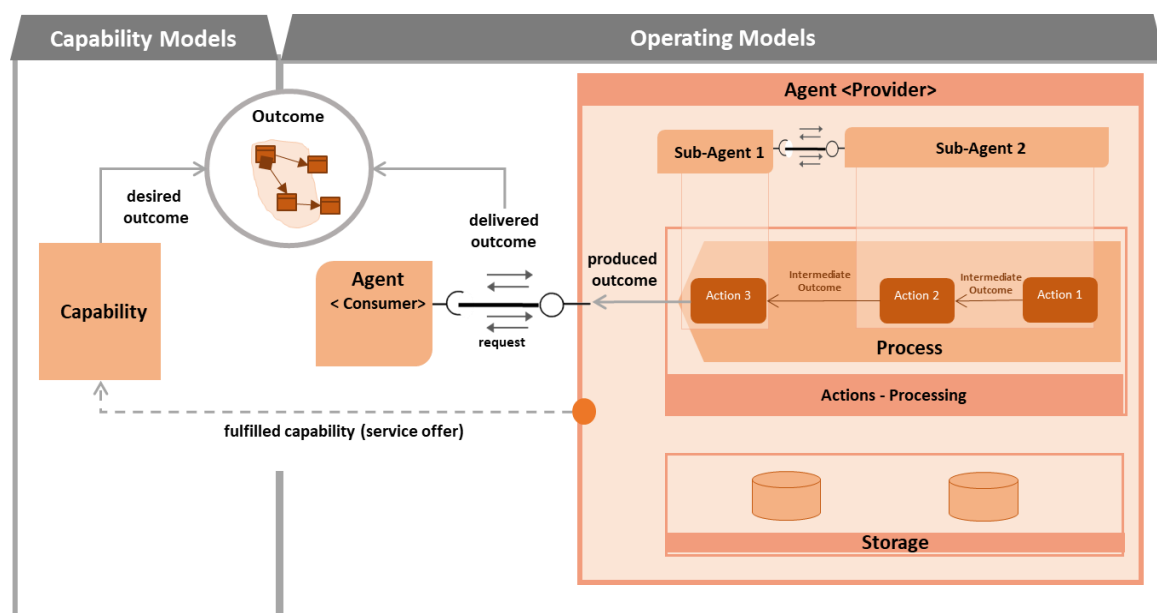


Figure 23 6 – Capability & Operating Model

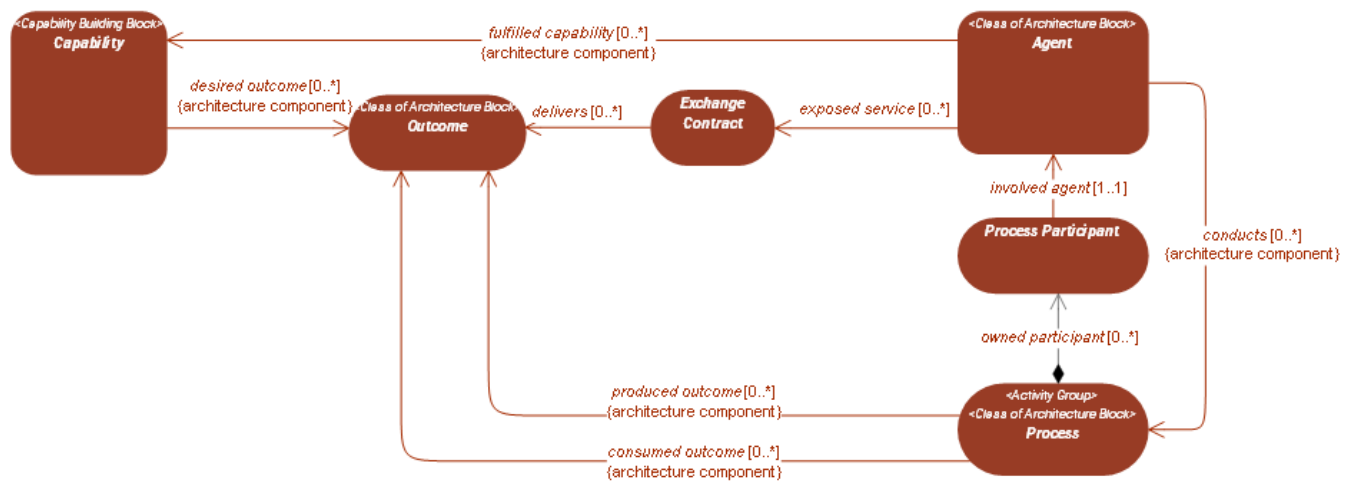


Figure 24 6 – Operating Model Pattern

 Operating Model Pattern	
Agent	Concept Type 1. An Architecture Block which is able to participate actively to a Process and/or to conduct a Process. (Hypernyms) Architecture Block (Hyponyms) Business Agent, Resource Agent (Component 1) agent component: A directed reference relationship between Agents and their component Agents. (Type) Agent (Presence) Optional (Cardinality) * (Hypernyms) architecture component (Hyponyms) resource agent component, component business function (Component 2) information store (Type) Information Area (Presence) Optional (Cardinality) * (Hypernyms) architecture component (Component 3) exposed service: An interaction process that occurs between Agents in order to deliver an Outcome. (Type) Exchange Contract (Presence) Optional (Cardinality) * (Hypernyms) architecture component (Hyponyms) exposed API, exposed business service (Component 4) fulfilled capability: Capabilities fulfilled by an Agent. (Type) Capability (Presence) Optional (Cardinality) * (Hypernyms) architecture component (Hyponyms) fulfilled business capability, fulfilled hardware functionality, fulfilled functionality (Component 5) performed process: A typical course of action intended to achieve an Outcome through the involvement of Agents (Type) Process (Presence) Optional (Cardinality) * (Hypernyms) architecture component (Hyponyms) performed resource process (Component 6) Capability Fulfillment (Type) Capability Fulfillment (Presence) Optional (Cardinality) *
Capability	Concept Type 1. Ability to deliver an Outcome .

	Capabilities are fulfilled by Agents. (Hypernyms) Capability Building Block (Hyponyms) Functionality, Skill, Business Capability, Technical Functionality, Hardware Functionality (Component 1) desired outcome: Result of a process conducted by an Agent . (Type) Outcome (Presence) Optional (Cardinality) * (Hypernyms) architecture component (Hyponyms) desired resource outcome, desired hardware outcome, desired technical outcome, desired business outcome
Exchange Contract	 Concept Type 1. An interaction process that occurs between Agents in order to deliver an Outcome. (Hypernyms) Architecture Block (Hyponyms) Resource Exchange Contract, Business Exchange Contract (Component 1) delivered outcome: Result of a process conducted by an Agent . (Type) Outcome (Presence) Optional (Cardinality) *
Outcome	 Concept Type 1. Result of a process conducted by an Agent . (Hypernyms) Architecture Block (Hyponyms) Hardware Work Product, Communication Format, Resource Outcome, Business Outcome
Process	 Concept Type 1. A typical course of action intended to achieve an Outcome through the involvement of Agents . (Hypernyms) Activity Group, Architecture Block (Hyponyms) Value Stream, Resource Process (Component 1) produced enterprise outcome: Outcome which is a result of the completion of a Process. (Type) Outcome (Presence) Optional (Cardinality) * (Hypernyms) architecture component (Hyponyms) produced enterprise outcome, produced resource outcome (Component 2) consumed enterprise outcome: Outcome which is required as an input of a Process. (Type) Outcome (Presence) Optional (Cardinality) * (Hypernyms) architecture component (Hyponyms) consumed resource outcome (Component 3) owned participant: Inclusion relationship that states that a Process is structurally composed of the involvement of Agents as Process Participants. (Type) Process Participant (Presence) Optional (Cardinality) * (Hypernyms) owned architecture component (Hyponyms) owned resource process participant, owned value stage
Process Participant	 Concept Type 1. Involvement of an Agent in a Process through which the Agent is assigned activities. (Hypernyms) Architecture Block Component (Hyponyms) Resource Process Participant, Value Stream Participant (Component 1) involved agent: Reference relationship that states which Agent is involved as a Process Participant. (Type) Agent (Presence) Always (Cardinality) 1 (Hypernyms) architecture block role

	(Hyponyms) Involved Business Agent, involved resource agent (Component 2) carried out activity: A reference relationship that states that a Process Participant carries out Process Activity(ies). (Type) Process Activity (Presence) Optional (Cardinality) * (Hyponyms) carried out stage, carried out resource activity
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4.4 Business Capabilities

BACM uses the term “business capabilities” to represent strategic capabilities.

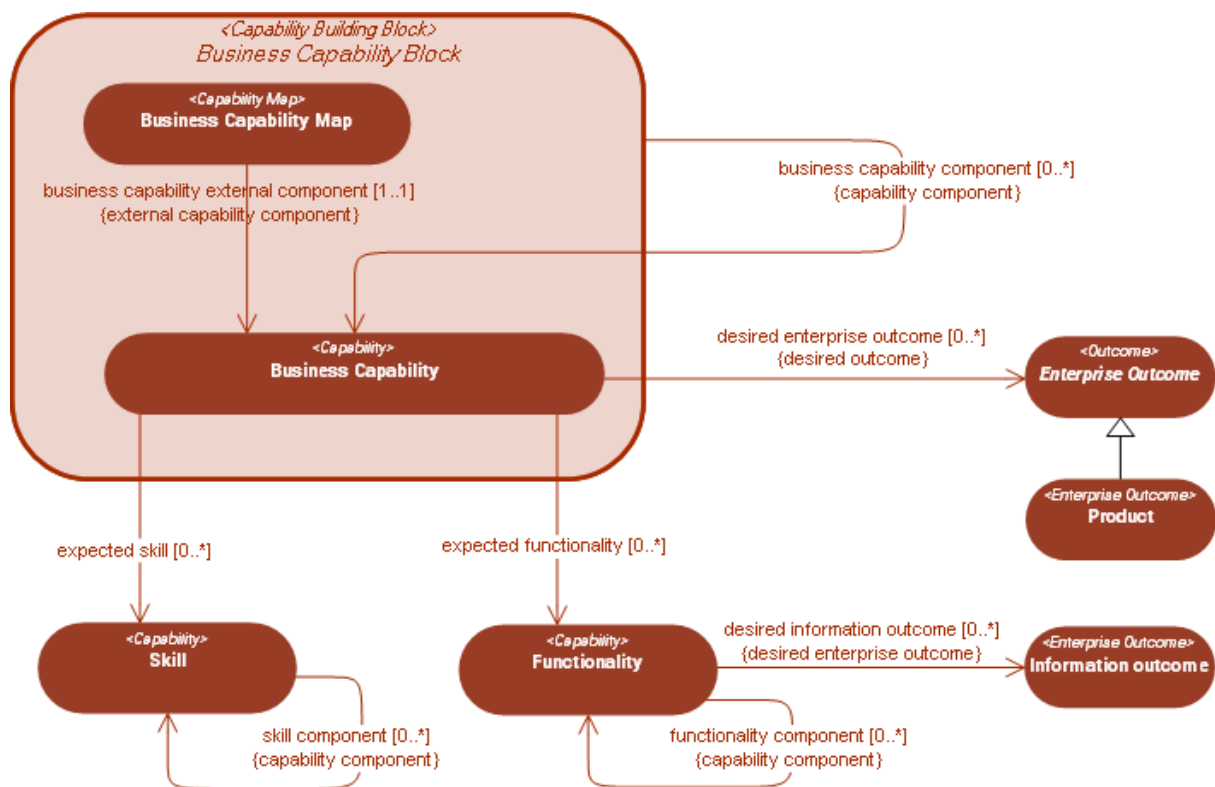


Figure 25 6 – Business Capability Meta-Model

Business Capability Meta-Model	
Business Capability	Concept Type 1. Strategic Capability of an Enterprise used to described its business model in terms of products & services it is aiming at delivering to its internal or external customer. A Business Capability is defined by its intended Business Outcomes and the kind performance (Measure Dimension) under which the delivry of these Business Outcomes of shall be proceeded. The actual performance value for a given by an enterprise at its difference stages is given by their exhibition (Exhibited Capability) and the stage level. (Hypernyms) Capability (Component 1) expected skill: Skill that is expected to be possessed by human resources contributing to implementing the business capability. (Type) Skill (Presence) Optional (Cardinality) * (Component 2) desired business outcome: Desired Business

	Outcome which shall result from the fulfillment of the Business Capability by the enterprise. (Type) Business Outcome (Presence) Optional (Cardinality) * (Hypernyms) desired outcome (Hyponyms) desired information outcome (Component 3) expected functionality: Functionality that is expected to be delivered by solutions implementing the business capability. (Type) Functionality (Presence) Optional (Cardinality) *
Business Capability Block	 Concept Type 1. Assembly of Business Capability(ies) . Sub-Types of Business Capability Block are either Business Capability Map or Business Capability. (Hypernyms) Capability Building Block (Component 1) business capability component: usiness capabilities in scope of a capability map. (Type) Business Capability (Presence) Optional (Cardinality) * (Hypernyms) capability component
Business Capability Map	 Concept Type 1. Top level assembly of Business Capability(ies) and their dependencies that, together, provide a business capability scope for an Enterprise Stage. (Hypernyms) Capability Map (Component 1) business capability external component: External Business Capability involved in an enterprise capability map. There express the Business Capability that the enterprise depends on but has delegated to partners. (Type) Business Capability (Presence) Always (Cardinality) 1 (Hypernyms) external capability component

4.5 Service & Product Model

To be done.

4.6 Business Operating Model

A business operating model is an abstract representation of how an enterprise operates across a range of domains in order to accomplish its missions as defined by its business model. It is organized as a triptych which expresses the way agents operate to deliver desired outcomes:

- Business Agent interact: service exchanges.
- Business Agents act: value streams.
- Business Agent store: memorization and storage of information : there are no flows without stores

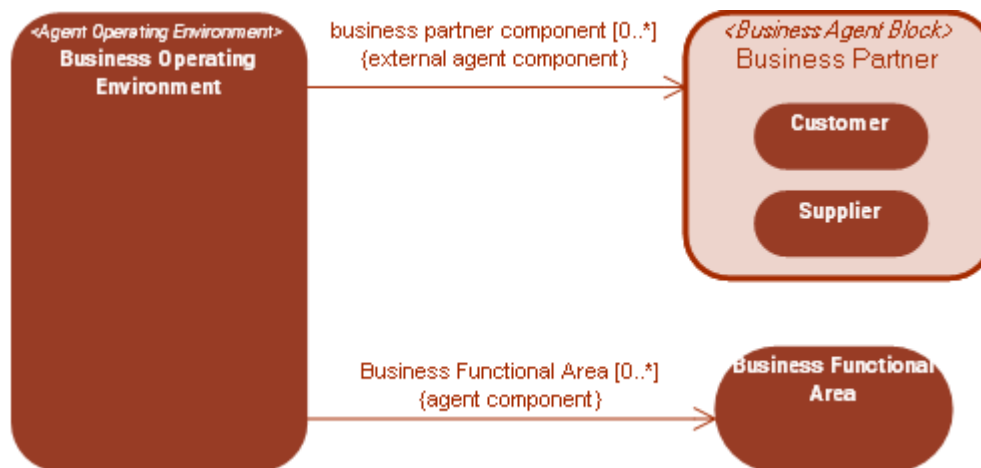


Figure 26 6 – Business Operating Environment Model

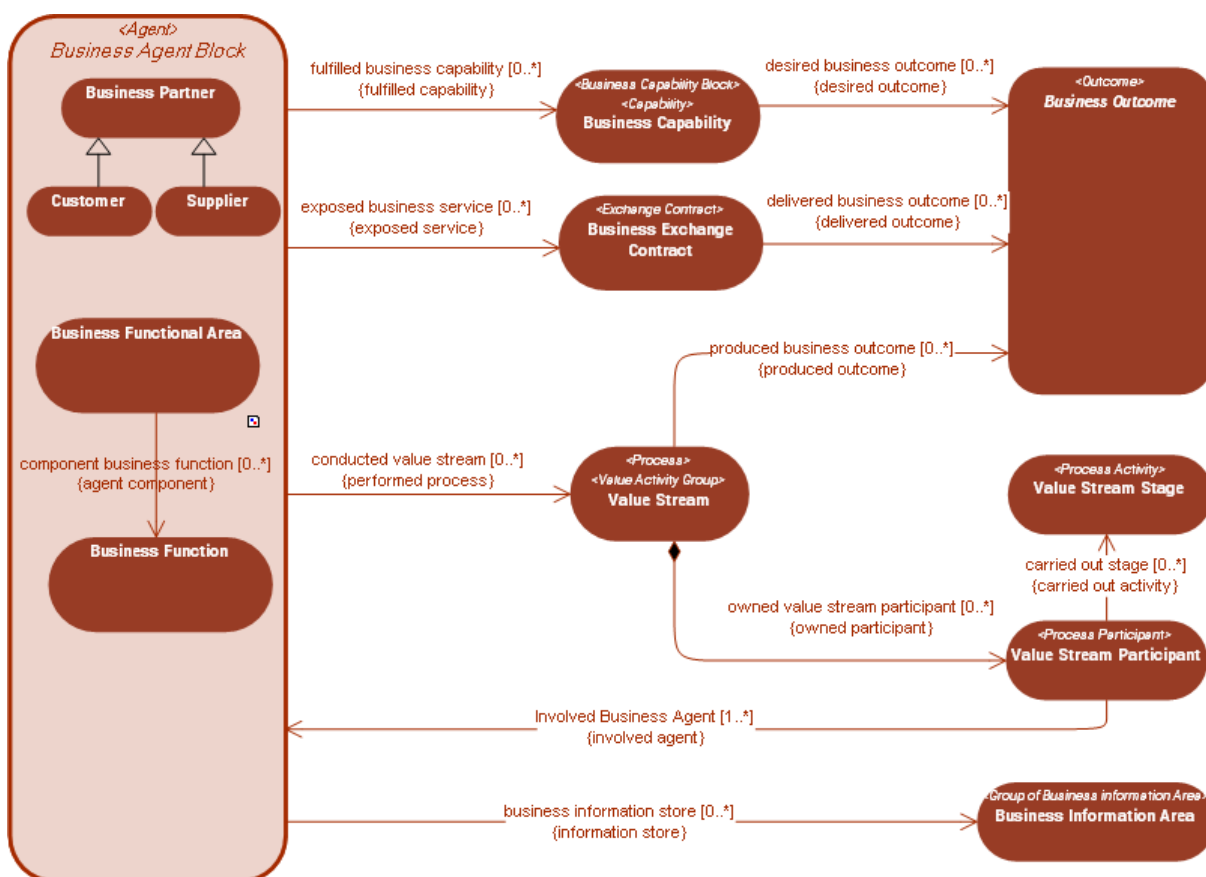


Figure 27 6 – Business Operating Model

Business Operating Model	
Business Agent Block	Concept Type 1. Assembly of business agent that, together, fulfill a mission. (Hypernyms) Agent (Hyponyms) Business Function, Business Partner, Business Functional Area (Component 1) exposed business service (Type) Business Exchange Contract (Presence) Optional (Cardinality) * (Hypernyms) exposed service (Hyponyms) exposed resource service (Component 2) fulfilled business capability: Business Capability

	fulfilled by a business agents by means of actions carried out by Business Function and interactions occurring between these Business Functions (Type) Business Capability (Presence) Optional (Cardinality) * (Hypernyms) fulfilled capability (Component 3) conducted value stream: Value Stream conducted by a business agent. (Type) Value Stream (Presence) Optional (Cardinality) * (Hypernyms) performed process (Component 4) business information store (Type) Business Information Area (Presence) Optional (Cardinality) * (Hypernyms) information store
Business Exchange Contract	 Concept Type 1. (Hypernyms) Exchange Contract (Hyponyms) Application Exchange Contract (Component 1) delivered business outcome: Business result of a Value Stream conducted by an Agent . (Type) Business Outcome (Presence) Optional (Cardinality) * (Hypernyms) delivered outcome (Component 2) delivered outcome: Information Outcome that is created by one or more Agents following some Process and that is communicated and consumed by other Agent . (Type) Information outcome (Presence) Always (Cardinality) * (Component 3) Exchange Node (Type) Exchange (Presence) Optional (Cardinality) *
Business Function	 Concept Type 1. Unit of the conceptual breakdown on enterprise responsibilities, used to shape its management structure in regard to how it produces, consumes, or process business outcomes: information, energy, materiel. Business functions specify competencies and functionalities required to achieve their duty. (Hypernyms) Business Agent Block (Component 1) required functionality: Functionality required by a business function in order to be able to operate. (Type) Functionality (Presence) Optional (Cardinality) * (Component 2) required skill: Skill required for the business function to be able to operate. (Type) Skill (Presence) Optional (Cardinality) *
Business Functional Area	 Concept Type 1. Assembly of interacting Business Functions that, together, delivery one of more business capabilities . Business Functional Areas, are the top level grouping of Business Functions at the level of the enterprise Business Operating Environment. (Hypernyms) Business Agent Block (Component 1) sub-business function: Composition of a Business Function which in a parent Business Functional Area . (Type) Business Function (Presence) Optional (Cardinality) * (Component 2) sub-business architecture: Composition of a sub Business Function Architecture in a parent Business Function

	Architecture . (Type) Business Functional Area (Presence) Optional (Cardinality) * (Component 3) component business function: Business Function being assembled by a Business Functional Area. (Type) Business Function (Presence) Optional (Cardinality) * (Hypernyms) agent component
Business Operating Environment	 Concept Type 1. Business environment in which the enterprise operate to fulfill its business capabilities and deliver expected value to to customers while interacting with its partners. (Hypernyms) Agent Operating Environment (Component 1) business partner component: A type of stakeholder that interacts with the enterprise in the context its business operating environment. Example : Private Sector Customer, Regulation Authority, (Type) Business Partner (Presence) Optional (Cardinality) * (Hypernyms) external agent component (Component 2) Business Functional Area: Assembly of interacting Business Functions that, together, delivery one of more business capabilities . Business Functional Areas, are the top level grouping of Business Functions at the level of the enterprise Business Operating Environment. (Type) Business Functional Area (Presence) Optional (Cardinality) * (Hypernyms) agent component
Business Partner	 Concept Type 1. A type of stakeholder that interacts with the enterprise in the context its business operating environment. Example : Private Sector Customer, Regulation Authority, (Hypernyms) Business Agent Block (Hyponyms) Customer, Supplier
Customer	 Concept Type 1. A Business Partner which is a customer of the enterprise. (Hypernyms) Business Partner
Supplier	 Concept Type 1. A Business Partner which is a supplier of the enterprise. (Hypernyms) Business Partner

4.7 Value Stream Model

The value stream model is a specialization of BPMN, aimed at describing value creation at the level of the business operating model. It describes how business functions act and interact in order to deliver business capabilities.

Each value stage produces and consumes outcomes to deliver the final outcome of the value stream. The various stages are carried out by business agents, usually at the level of business functions. Each stage may require a capability for its achievement. These required capabilities can be business capabilities, functionalities, or skills.

It is up to the various resource operating model (people, software, hardware) to specify how value stages and required capability are carried out and implemented.

The split of value streams into value stage shall be carefully synchronized with the split of business functions so that the operating triptych is made consistent (see business operating model).

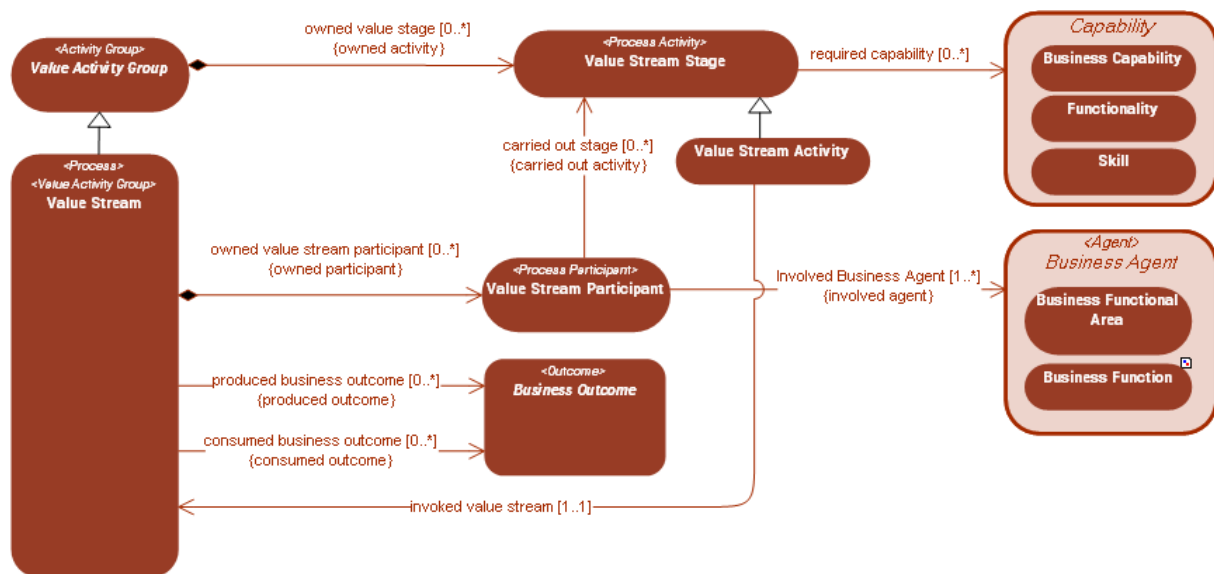


Figure 28 6 – Value Stream Meta-Model

Value Stream MetaModel	
Value Activity Group	Concept Type 1. A group of Value Stream Stage(s). (Hypernyms) Activity Group (Hyponyms) Value Stream (Component 1) owned value stage: An inclusion relationships that stated that an Value Activity Group is made of Value Stream Stages. (Type) Value Stream Stage (Presence) Optional (Cardinality) * (Hypernyms) owned activity
Value Stream	Concept Type 1. A process which is conducted by Business Agent Blocks (Business Function or Business Functional Area) and which describes how the enterprise operates at the conceptual level. (Hypernyms) Value Activity Group, Process (Component 1) produced business outcome: Business Outcome produced by a value stream, conducted under the responsibility of business agent. (Type) Business Outcome (Presence) Optional (Cardinality) * (Hypernyms) produced outcome (Component 2) owned value stream participant: Role of a business agent in a value stream. (Type) Value Stream Participant (Presence) Optional (Cardinality) * (Hypernyms) owned participant (Component 3) consumed business outcome: Business result of a Value Stream conducted by an Agent . (Type) Business Outcome (Presence) Optional (Cardinality) * (Hypernyms) consumed outcome
Value Stream Activity	Concept Type 1. A Value Stream Stage which is the invocation of another Value Stream (Hypernyms) Value Stream Stage

	(Component 1) invoked value stream: Reference relationship that states which Value Stream is invoked as part of a Value Stream Activity. (Type) Value Stream (Presence) Always (Cardinality) 1
Value Stream Participant	Concept Type 1. Role of a business agent in a value stream. (Hypernyms) Process Participant (Component 1) carried out stage: A reference relationship that states that a Value Stream Participant carries out Value Stream Stages. (Type) Value Stream Stage (Presence) Optional (Cardinality) * (Hypernyms) carried out activity (Component 2) Involved Business Agent: Reference relationship that states which business agent is involved as a Value Stream Participant. (Type) Business Agent Block (Presence) Always (Cardinality) * (Hypernyms) involved agent
Value Stream Stage	Concept Type 1. An added value step in a Value Stream (Hypernyms) Process Activity (Hyponyms) Value Stream Activity (Component 1) required capability: A reference relationship that states that a Value Stream Stage requeries Capability(ies) for its execution. These Capability(ies) can be Business Capability, Functionality or Skill. (Type) Capability (Presence) Optional (Cardinality) *

4.8 Resource Operating Model

The resource operating model is an abstract pattern that serves as a hook for all other resource meta-model that

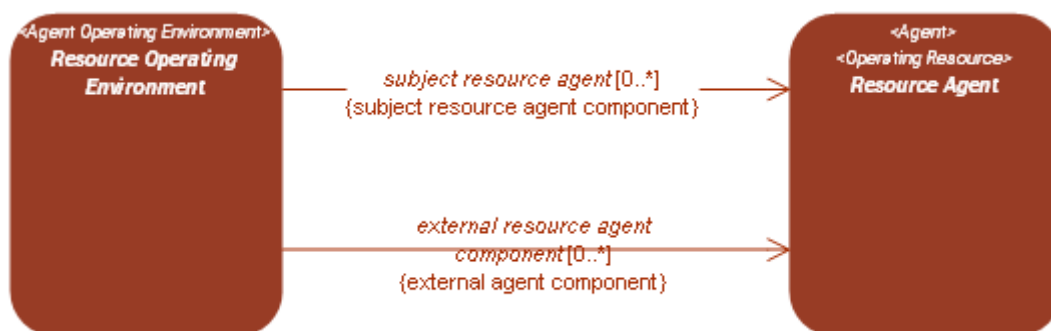


Figure 29 6 – Resource Operating Environment Pattern

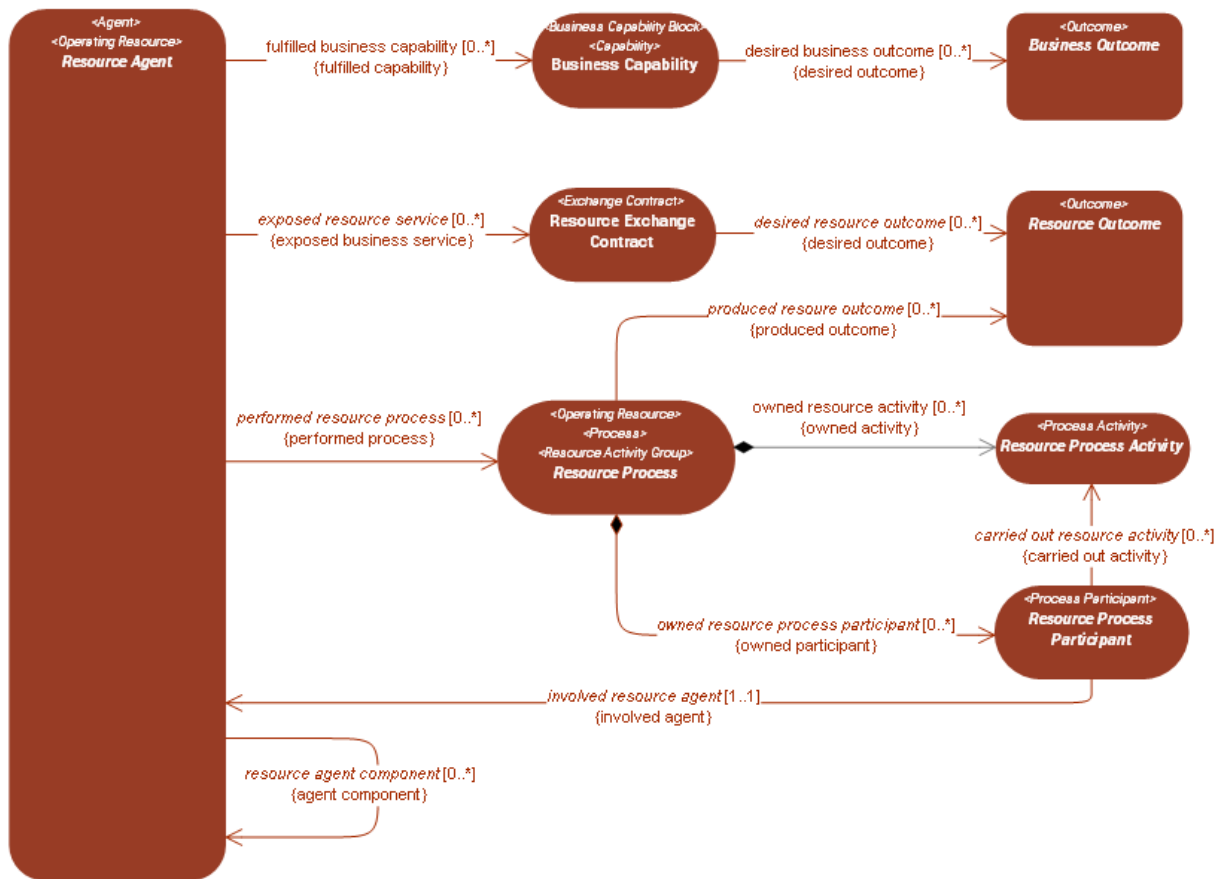


Figure 30 6 – Resource Operating Agent Pattern

4.9 Resource Functionality Models

Resource Functionalities are capabilities of physical resources that are required from the business operating model. They include capabilities to produce information outcomes (functionalities), capabilities to produce hardware outcomes (hardware functionalities) and human skills.

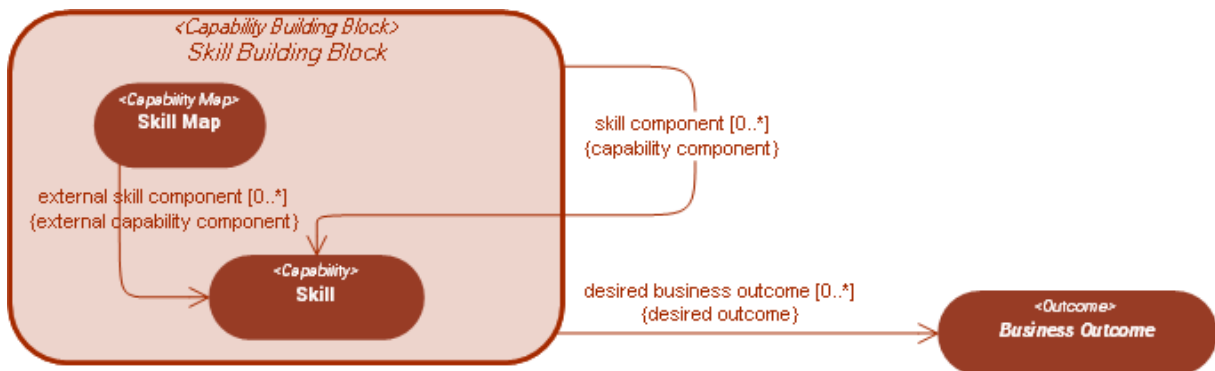


Figure 31 6 – Skill Meta-Model

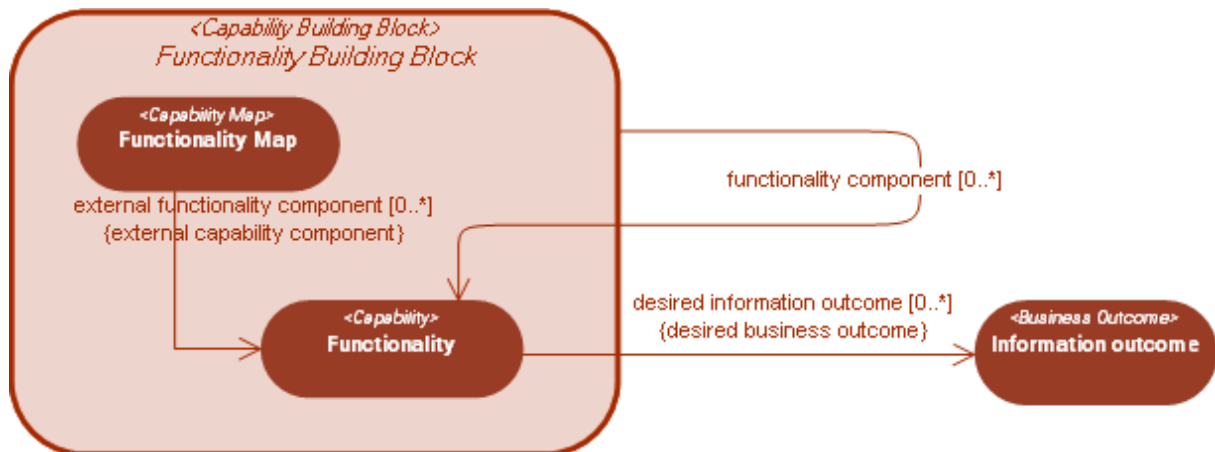


Figure 32.6 – Functionality Meta-Model

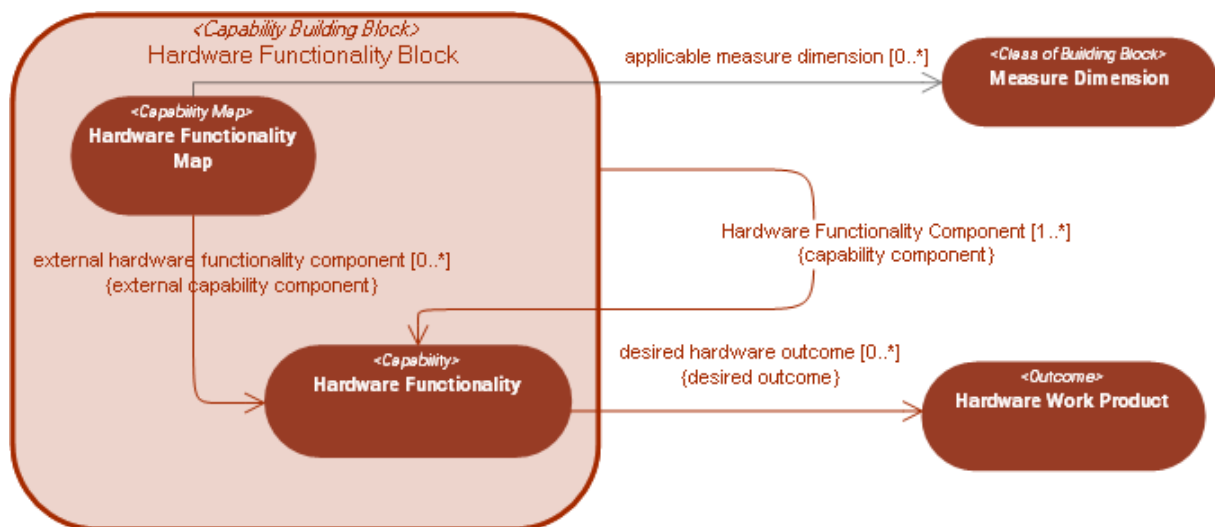


Figure 33.6 – Hardware Capability Meta-Model

4.10 Organization Model

The organization mode is a specific kind of resource operating model that deals with human resources and the split of responsibility among business units.

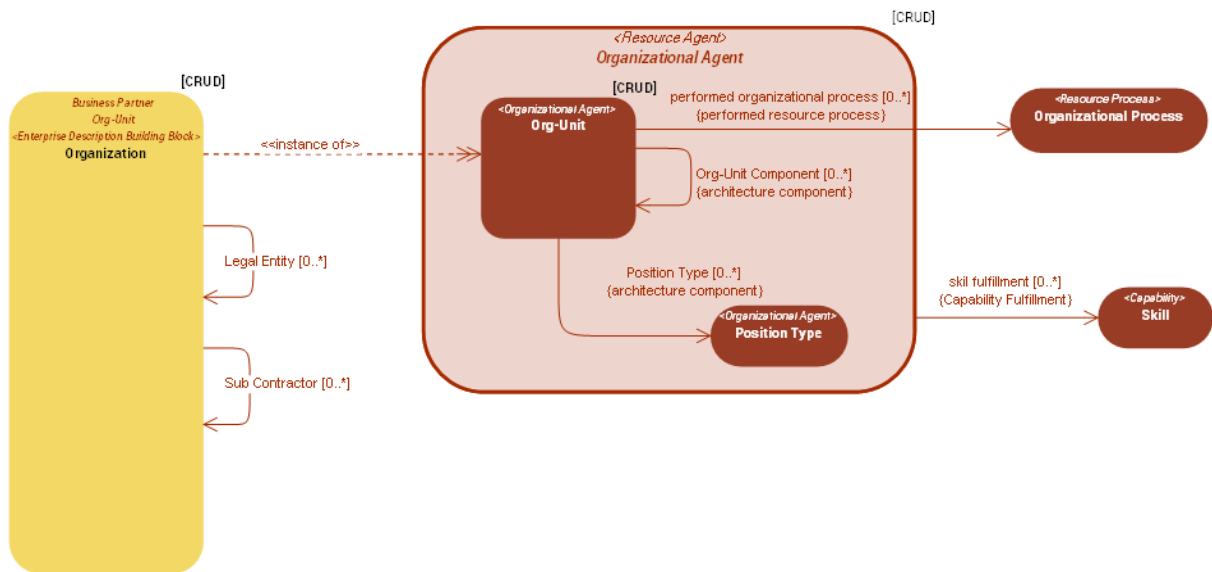


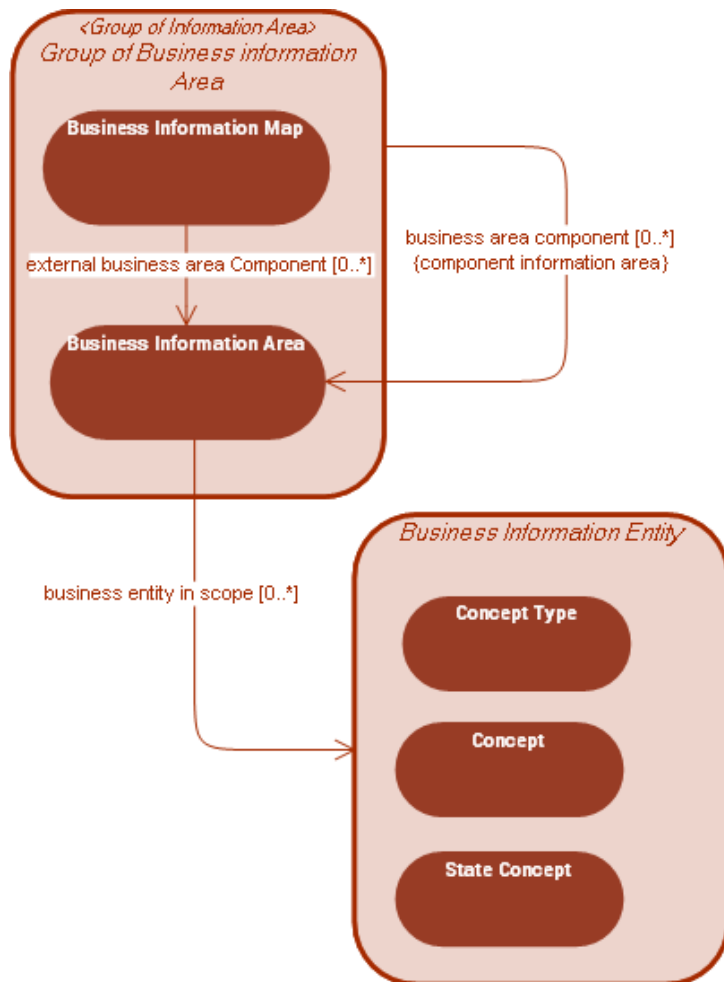
Figure 34 6 – Organization Meta-Model

4.11 Information Model

Business Concepts map are used to describe enterprise business information.

As stated by the upper ontology all relationships are directed. The BACM information model includes a temporal view of information through the embedding of states in business information concepts.

To be finalized.



4.12 Strategy Model

BACM strategy model is grounded on OMG BMM to which two main aspects are added: strategic capabilities and a temporal aspect of the Enterprise.

BMM is embedded in a 4D framework which looks at the Enterprise as an overarching view of strategic capabilities overtime, along with their target operating models.

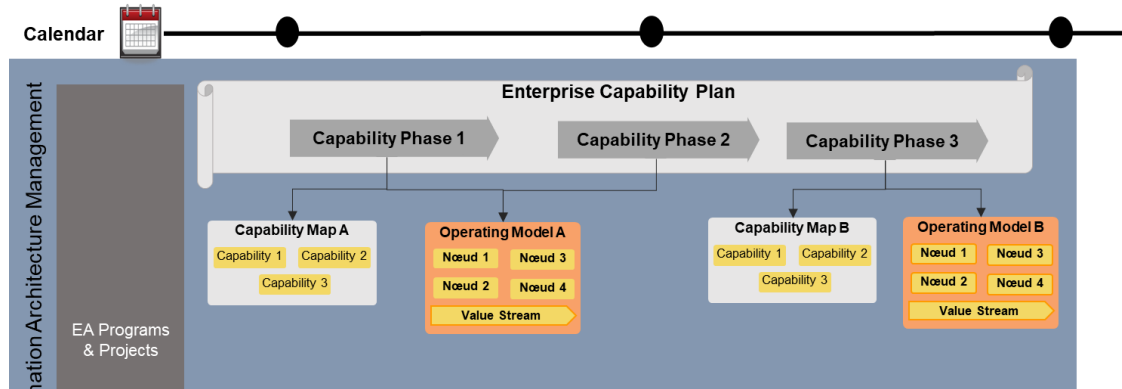


Figure 35 6 – Overarching of strategic capabilities & operating model over time

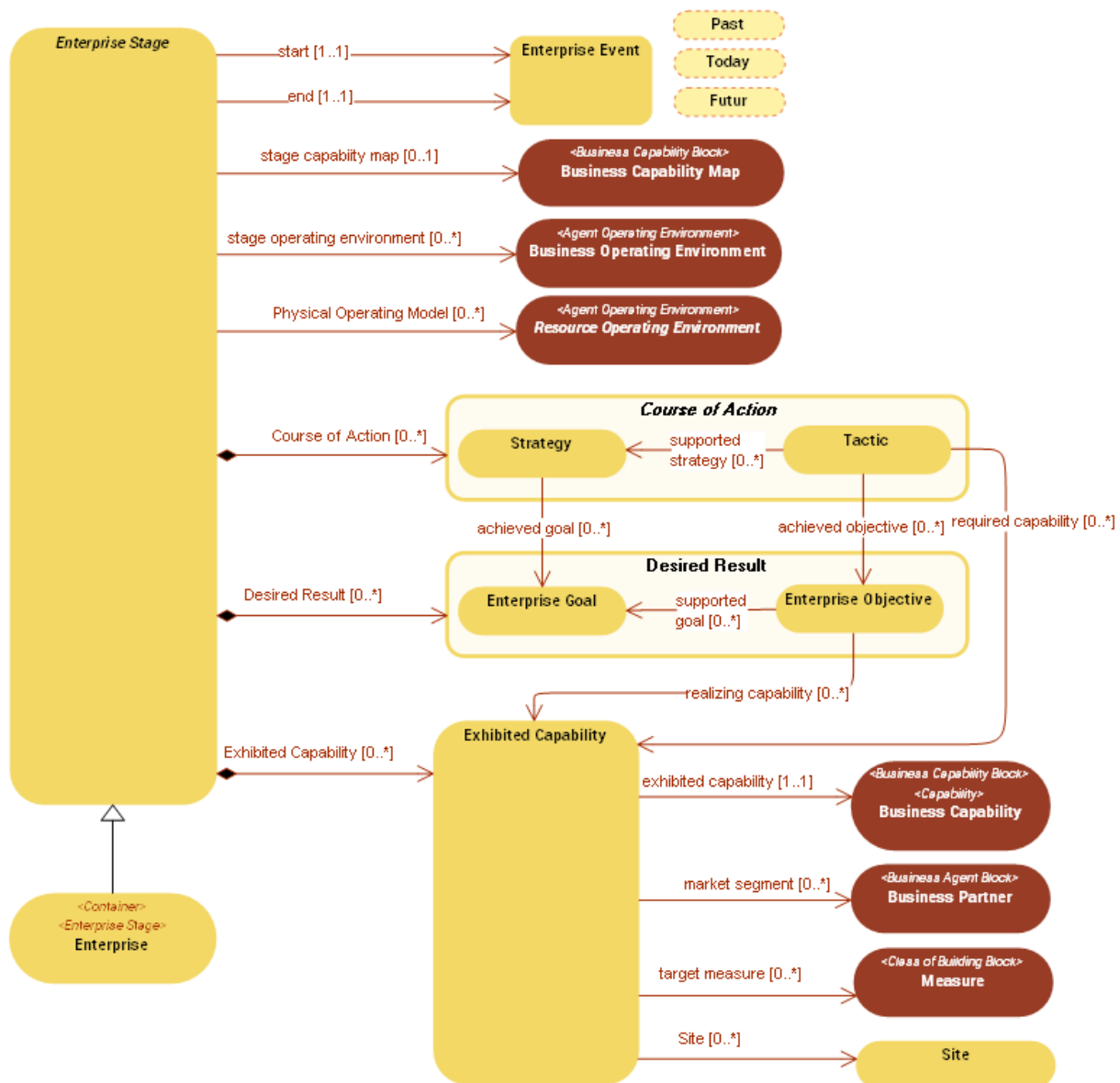


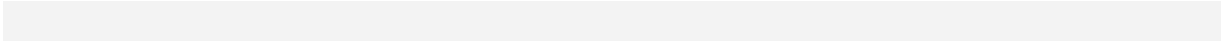
Figure 36 6 – Enterprise & Strategy MetaModel

Strategic Roadmapping	
Course of Action	Concept 1. An Plan recognized by an enterprise as being essential to achieving its goals - i.e. a strategic specification of what the enterprise does. In other words, a Course of Action channels efforts towards Desired Results. Business Capabilities might be required by an Enterprise to conduct its Courses of Action. (Hyponyms) Strategy, Tactic
Enterprise	Concept 1. A stage of the enterprise that that represents its whole existence. (Hypernyms) Enterprise Stage, Container
Enterprise Event	Concept 1. Something that happens at a given place and time and that marks the end of one Enterprise Stage and possibly the beginning of another. (Synonyms) Milestone
Enterprise Goal	Concept 1. A Goal is a statement about a state or condition of the enterprise to be brought about or sustained through appropriate Means. A Goal

	amplifies a Vision; that is, it indicates what must be satisfied on a continuing basis to effectively attain the Vision. A Goal should be narrow - focused enough that it can be quantified by Objectives. A Vision, in contrast, is too broad or grand for it to be specifically measured directly by Objectives. However, determining whether a statement is a Vision or a Goal is often impossible without in-depth knowledge of the context and intent of the business planners. (From BMM). (Hypernyms) Desired Result
Enterprise Objective	 Concept 1. An Objective is a statement of an attainable, time-targeted, and measurable target that the enterprise seeks to meet in order to achieve its Goals. · Attainable - It is self-evident that Objectives should be attainable. If they are not, the business plans are unrealistic and will likely fail. · Time-targeted - All Objectives should be time-targeted. This means that either an absolute time frame (e.g., "by January 1, 2007") or relative time frame (e.g., "within two years") should be included in each Objective. This timeframe indicates when the Objective is to be met. · Measurable - Objectives should be measurable. This means they must include some explicit criteria for determining whether the Objective is being met in practice. This criteria may be fairly exacting (for example, "on-time 95% of the time"). At the very minimum, the criteria must provide a basis for making a "yes or no" determination (e.g., "up and running"). Such criteria may be the basis for certain Business Rules, created specifically to compute or derive the relevant evaluation. This understanding of 'Objective' is consistent with the industry's popular "SMART" criteria that an Objective be: Specific, Measurable, Attainable, Relevant, and Time-Based. (Hypernyms) Desired Result (Component 1) realizing capability: Capabilities, which for a specific phase, along with their associated performance characteristic contribute to the achievement of desired results. (Type) Exhibited Capability (Presence) Optional (Cardinality) *
Enterprise Stage	 Concept 1. A past, current or future state of the enterprise. Each stage represents an undertaking comprising a purposeful set of activities whose primary purpose is focused on achieving a set of clearly defined objectives that may transcend organisational boundaries and consequently require integrated team working under the direction of a governing body of stakeholders. (Hyponyms) Enterprise (Component 1) Physical Operating Model: Assertion that an enterprise stage has a set of reference Business Operating Environments that specify its physical operating structures. Business Operating Environments can be organizational operating environments, software operating environments or infrastructure operating environments. (Type) Resource Operating Environment (Presence) Optional (Cardinality) * (Component 2) stage capability map: Assertion that an enterprise stage has a scoped set of business capabilities. (Type) Business Capability Map (Presence) Optional (Cardinality) 1 (Component 3) stage operating environment: Assertion that an enterprise stage has a reference Business Operating Environment (Type) Business Operating Environment (Presence) Optional (Cardinality) * (Component 4) end: End event of an enterprise stage. (Type) Enterprise Event (Presence) Always (Cardinality) 1

	(Component 5) Exhibited Capability: Capability that is exhibited by an Enterprise Stage with quantified measure and potential geopolitical scope for a market segment. (Type) Exhibited Capability (Presence) Optional (Cardinality) * (Component 6) start: Start event an an enterprise stage. (Type) Enterprise Event (Presence) Always (Cardinality) 1 (Component 7) Course of Action: Strategic actions taken during the course of an Enterprise Stage to achieved it Ends. (Type) Course of Action (Presence) Optional (Cardinality) * (Component 8) Desired Result: Strategic Ends of an Enterprise Stage. (Type) Desired Result (Presence) Optional (Cardinality) *
Exhibited Capability	 Concept 1. Capability that is exhibited by an Enterprise Stage with quantified measure and potential geopolitical scope for a market segment. (Component 1) exhibited capability: Business Capability being exhibited by an Enterprise . (Type) Business Capability (Presence) Always (Cardinality) 1 (Component 2) market segment: An identifiable group of individuals, families, businesses, or organizations that benefits from the effect of the exhibited capability. (from http://www.businessdictionary.com). (Type) Business Partner (Presence) Optional (Cardinality) * (Component 3) target measure: target Measures for the the exhibited capability. (Type) Measure (Presence) Optional (Cardinality) * (Component 4) Site: Geopolitical location where the capability is exhibited. (Type) Site (Presence) Optional (Cardinality) *
Milestone	 Concept 1. Something that happens at a given place and time and that marks the end of one Enterprise Stage and possibly the beginning of another. (Synonyms) Enterprise Event
Strategy	 Concept 1. A Strategy is one component of the plan for the Mission. A Strategy represents the essential Course of Action to achieve Ends (Goals in particular). A Strategy usually channels efforts towards those Goals. A Strategy is more than simply a resource, skill, or competency that the enterprise can call upon; rather, a Strategy is accepted by the enterprise as the right approach to achieve its Goals, given the environmental constraints and risks. (From BMM). (Hypernyms) Course of Action (Component 1) achieved goal: Enterprise Goals that the Strategy is aiming at achieving. (Type) Enterprise Goal (Presence) Optional (Cardinality) *
Tactic	 Concept 1. A Tactic is a Course of Action that represents part of the detailing of Strategies. Tactics generally channel efforts towards Objectives.

	(From BMM) (Hypernyms) Course of Action (Component 1) required capability: Exhibited Capability that is required to enable a Course of Action to achieve its Desired Result . (Type) Exhibited Capability (Presence) Optional (Cardinality) * (Component 2) achieved objective: Enterprise Objective that the Tactic is aiming at achieving. (Type) Enterprise Objective (Presence) Optional (Cardinality) *
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4.13 Transformation Model

The transformation model described project activities that lead from past enterprise phases to future enterprise phases. Project/sprint deliverables apply changes (create/update/remove) to physical operating build blocks that contribute to targeted enterprise operating models.

Capability planning is the preferred method to ensure transformation and agility at scale. It encompasses three distinct time horizons:

- The capability plan which defines expected capabilities over time along with their prescribed architecture solutions (see strategy model).
- The life-cycles of architecture solution building blocks: asset catalogs.
- Transformation increments delivering MVP.

The agility at scale capability approach consists in the design and governance of these three time horizons required to conduct enterprise transformations. Continuous architecture ensures that transformations follow of sustainable architecture principles overtime.

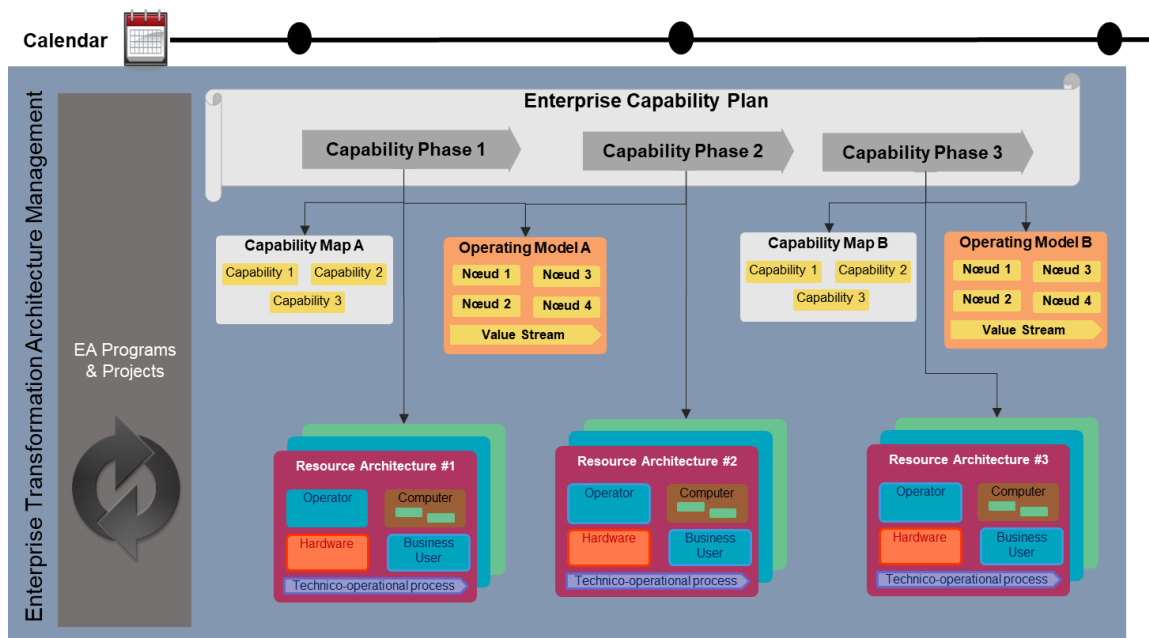


Figure 37 6 – Transformation & solution overarching

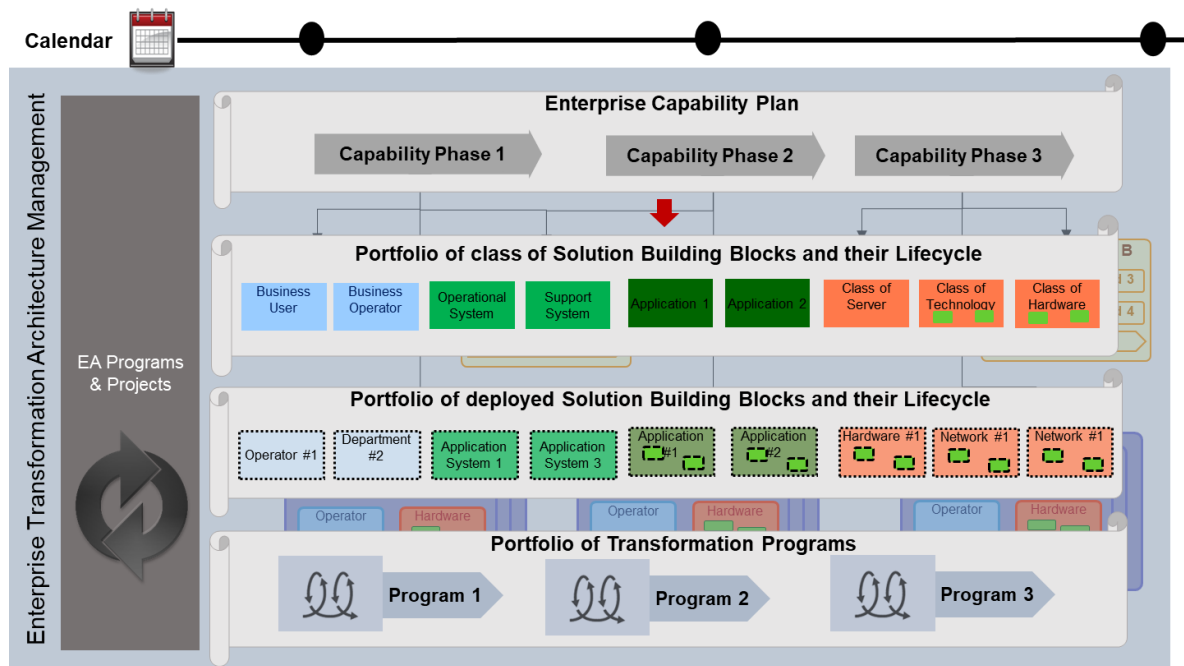


Figure 38.6 – Transformation time horizons

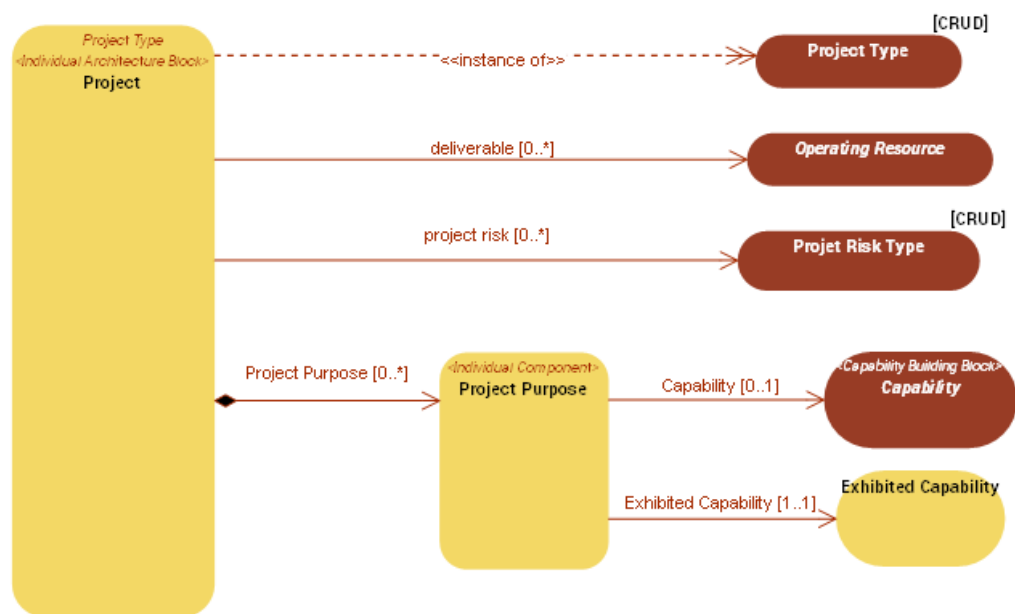


Figure 39.6 – Transformation Meta-Model

5 Compliance Levels

The following levels of compliance are defined for this specification.

5.1 PIM Compliance

An implementation is PIM compliant if it implements the complete PIM model in a specific technology.

5.2 Other?

Appendix A: Glossary

5.3 The BACM GRID

EA Stakeholders 	Object Kind → ↓ EA Layer	Enterprise Description Blocks								Category Building Blocks			
		Capability Building Blocks	Outcome	Operating Building Blocks			Information Building Block	Information Container	Policy Building Blocks	Measures	Categories		
				Agent									
				Service Exchange	Process	Store							
Business Architect Strategic Planner	Business Layer	Business Capability	Business Outcome	Business Function			«Business Information» Concept Concept State	Subject Area	Business Rules				
				Business Exchange	Value Stream	Business Store							
Business Analyst	Organization Layer	Business Capability Skill	Business Outcome	Org-Unit			Class	Package	Operational Rules				
				Organization Exchange	Business Process	Logical Data Store	Data Entity						
	Other Resources												
	Enterprise	Exhibited Capabilities	Strategy & Planned Architecture							Planned Performance			
		Projects	Planning of Enterprise Transformations										

5.4 Table of figure

Figure 1 – BACM Scope: Motivation/Capability/Operating Model – All over time	5
Figure 2 – Composite structures and modularity.....	7
Figure 3 – 4D approach of identify over time	7
Figure 4 – Knife over time	9
Figure 5 – Traditional systemic viewpoints	10
Figure 6 – Goals & Purposes (Capabilities).....	11
Figure 7 – IT Service in OpenGroup W161 whitepaper	12
Figure 8 – The operating triptych and capabilities.....	12
Figure 9 – System & Purpose Approach	13
Figure 10 - Russell Ackoff classification of systems.....	13
Figure 11 – The Enterprise as an evolving system defining its own purposes overtime	14
Figure 12 - Various kinds of gaps and corresponding alignments.....	14
Figure 13 - BACM simplified GRID	15
Figure 14 6 – Business Model Canvas	17
Figure 15 – Value Proposition Canvas	17
Figure 16 – BACM Concept Map	18
Figure 17 6 –BACM Concept Map	20
Figure 18 6 – Syntax Meta-Model	20
Figure 19 6 – Individual / Class Meta-Model.....	22
Figure 20 6 – Architecture Block Pattern	23
Figure 21 6 – Capability Pattern	25
Figure 22 6 – Process Pattern.....	26

Figure 23 6 – Capability & Operating Model.....	28
Figure 24 6 – Operating Model Pattern	29
Figure 25 6 – Business Capability Meta-Model	31
Figure 26 6 – Business Operating Environment Model.....	33
Figure 27 6 – Business Operating Model.....	33
Figure 28 6 – Value Stream Meta-Model	36
Figure 29 6 – Resource Operating Environment Pattern	37
Figure 30 6 – Resource Operating Agent Pattern.....	38
Figure 31 6 – Skill Meta-Model	38
Figure 32 6 – Functionality Meta-Model.....	39
Figure 33 6 – Hardware Capability Meta-Model	39
Figure 34 6 – Organization Meta-Model	40
Figure 35 6 – Overarching of strategic capabilities & operating model over time	42
Figure 36 6 – Enterprise & Strategy MetaModel	43
Figure 37 6 – Transformation & solution overarching	47
Figure 38 6 – Transformation time horizons.....	48
Figure 39 6 – Transformation Meta-Model.....	48